AI Epistemic Risks: Emerging Mechanisms & Evidence

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Executive Summary

Humanity's ability to know, reason, judge, and act well is the foundation of the institutions that enable scientific progress, democratic governance, crisis response, and the management of AI itself. This paper argues that AI advances pose serious risks to that foundation. We call these epistemic risks—threats to humanity's collective capacity to know things accurately, reason well, form beliefs, and maintain a healthy information environment.

Epistemic risk is not misinformation by another name. It runs deeper: it arises from AI's integration into the very infrastructure through which we think, form beliefs, and make sense of the world together. Across fields including technical machine learning, AI safety, human-computer interaction, philosophy and ethics, cognitive neuroscience, and education, researchers are studying different fragments of a shared problem. This paper strives to connect those fragments, analyze how their harms amplify one another to pose systemic and potentially catastrophic risks, and help surface cross-cutting solutions that no single research community has yet found on its own.

Epistemic risks demand particular urgency because their harms compound and are self-concealing. The longer they go unaddressed, the less capacity remains to address them: as individuals and institutions surrender epistemic autonomy, the very capabilities needed to recognize and respond to these risks are degraded. The paper identifies and focuses on three primary mechanisms by which AI can cause systemic epistemic decline, and traces how each can exacerbate other risks humanity faces:

- **Persuasion and manipulation.** AI systems are already highly persuasive and becoming more so. These capabilities are being actively used by state and non-state actors for political influence, radicalization, and economic manipulation. Harm can also arise without any malicious intent: sycophantic model behavior reinforces users' existing beliefs, increases attitude extremity, and erodes the willingness to seek disconfirming information. As AI becomes more personalized, capable, and widely deployed as a persistent interlocutor rather than a passive tool, the potential for harm scales accordingly, including at the level of decision-making by collective institutions. This is especially consequential given the need for global coordination on how advanced AI is governed.
- **Cognitive offloading.** AI's natural language interface enables a qualitatively different form of cognitive delegation than prior technologies. While search engines have affected information recall, modern AI intervenes at deeper layers: belief formation, explanation-making, and reasoning itself. Early evidence is accumulating: software engineers showed reduced debugging ability after AI-assisted coding (Shen and Tamkin, 2026); EEG studies found reduced neural connectivity during AI-assisted writing (Kosmyna et al., 2025). This risk is self-perpetuating: as individuals lose trust in their own reasoning, they increasingly default to what the literature calls "cognitive surrender," delegating higher-order thinking to AI. Without intervention, the result may be a systematic hollowing of cognitive resilience at both individual and societal levels.
- **Feedback loops and lock-in.** Human-AI and AI-AI feedback loops are narrowing the epistemic space from which humans and AI systems draw, resulting in multiple potential harms.[1] Evidence for homogenization is already documented: AI assistance has boosted individual scientific output while collectively narrowing research focus; AI-assisted brainstorming produces fewer unique ideas than human-only groups; LLM linguistic patterns have measurably entered human spoken and written communication. Depending on future platform and ecosystem choices, these loops may also amplify the echo-chamber dynamics beyond those documented in recommender systems. Epistemic "lock-in" is a hypothesized endpoint: a widespread loss of epistemic mobility in which society becomes trapped in self-referential cycles, resistant to corrective signals and unable to break toward new knowledge or value trajectories. If it occurs, reversal will be extremely difficult.
- These mechanisms are not exhaustive, and they amplify each other. People whose critical thinking has been dulled by cognitive offloading are more susceptible to AI-driven manipulation. Persuasive influence and

[1] Feedback loops are systems where the output of an action is fed back into the system as input, influencing future actions (Simon Fraser University, 2025); they are a key feature of complex systems (Ormand, Carol, 2025). In the context of our paper, we use them to refer to recursive cycles where AI-generated outputs—distributed between humans, between humans and AI, and between AI agents—influence subsequent inputs across the information ecosystem.

feedback loops together can exacerbate both homogenization and fragmentation of epistemic worlds, further constraining the space of ideas people encounter and take seriously.

Signs of epistemic harm are already emerging in specific domains, though systemic shifts remain difficult to measure and attribute. The technical drivers underpinning these mechanisms are present in today's systems and advancing. Some effects, including homogenization of research and linguistic outputs, professional skill degradation, reduced neural engagement during AI-assisted tasks, and altered belief formation from sycophancy, have been empirically documented. Measurement is nonetheless likely to only partially capture true impact: studies designed to isolate effect sizes may overestimate some effects, while ecosystem-level feedback dynamics are inherently difficult for individual studies to capture. These risks manifest gradually rather than acutely, and people who have lost epistemic capabilities may not recognize or report it. In addition, the gap between AI deployment speed and study timelines compounds the measurement problem structurally.

AI could be an unprecedented lever for improving epistemics, but this will not happen by default. Existing structural incentives, including engagement-driven optimization, competitive deployment pressure, and short-term productivity gains, actively push in the opposite direction. AI could strengthen epistemic capacity, improving humanity's ability to understand and manage AI risks as well as other challenges. Realizing that potential requires proactive, coordinated effort across governments, companies, and societal institutions.

Humanity's ability to handle AI risks and other major threats may depend on how epistemically resilient our institutions are. We highlight promising solutions, organized by where they intervene:

- **Within AI systems:** Epistemic-diversity-aware training objectives and pluralistic alignment approaches can counter homogenization; anti-sycophancy fine-tuning reduces manipulative reinforcement of user beliefs; epistemic fidelity requirements, designing outputs to be falsifiable and to surface uncertainty, create correction mechanisms that purely generative systems lack; maintaining real human data in training pipelines helps prevent model collapse.
- **In human-AI interaction:** Interface design can introduce deliberate cognitive friction to disrupt excess offloading and sycophancy; interaction patterns that require generation before comprehension have been shown to preserve cognitive skill formation; epistemic backstops that help users identify rhetorical manipulation, and bridging tools that support productive cross-group deliberation, offer paths toward AI that strengthens rather than weakens epistemic autonomy.
- **In human institutions:** Education reform should function as epistemic inoculation, cultivating specific skepticism toward AI failure modes rather than just generic media literacy, and institutions should identify which cognitive tasks must remain non-delegable. Regulatory frameworks should prioritize research access to platform data for epistemic monitoring. Scientific institutions must invest in metascience and interdisciplinary bridging to resist the monoculture dynamics already emerging in peer review and academic publishing.
- **Underlying incentive problems:** None of the above will be durable without addressing the institutional and commercial incentives that reward cognitive dependency, engagement over truthfulness, and speed over epistemic safety. Existing AI governance frameworks provide partial footholds, but epistemic risks as a class lack the threat models, response protocols, and international coordination infrastructure that risks like CBRN (Chemical, Biological, Radiological, and Nuclear) already have. Building that infrastructure is an urgent task.

This paper advances a risk-based argument, not a prediction. The mechanisms of AI-facilitated epistemic decline are plausible, grounded in how today's systems actually work, already partially evident in measured data, and non-negligibly likely to produce catastrophic outcomes at scale if current trajectories continue. The paper does not claim that AI must necessarily cause epistemic decline, that catastrophic outcomes are inevitable, or that AI cannot be built and used in better ways. It does claim that once epistemic degradation reaches certain thresholds, the individuals and institutions needed to address it may be too compromised to do so. The time to act is now, before our capacity to respond is itself lost.

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"Sure, I can help with that" — AI.

1. Introduction

Humanity's greatest problems are often, at their core, problems of collective decision-making. Despite remarkable achievements in science, medicine, and governance, we have struggled to effectively address civilizational challenges such as pandemics, climate change, and nuclear proliferation. Much of the difficulty stems from failures of reasoning, value disagreements, belief formation and shared sense-making, which in turn drive conflict and institutional breakdown (Haidt, 2012; Bar-Tal, 2013; Finkel et al., 2020; McCoy and Somer, 2019; Piazza, 2023).

Part of what makes many problems so difficult is that we are epistemically interdependent. Good epistemics, at both the individual and collective level, means being able to accurately form and update beliefs, reason independently, evaluate evidence, resist manipulation, and contribute to the shared markers of credibility that allow communities to distinguish expertise from shallow understanding and reliable knowledge from noise. It is not simply a matter of having access to correct information: it requires the cognitive capacity and social infrastructure to process, question, and build on that information together. Our ability to know and judge well depends on others doing the same (Kitcher, 1990). Good science depends on distributed communities of researchers checking and building on each other's work. Democratic institutions need to reflect individuals' autonomous decisions (Susser et al., 2019), and effective governance needs citizens who can evaluate evidence and resist manipulation of their political views (Susser et al., 2019).

These interdependencies are not incidental: society runs on the basis of division of cognitive labor, where laypeople rely on experts, institutions, and shared markers of credibility. The danger, therefore, is not interdependence itself, but the breakdown of these signposts (Rabb et al., 2019). And as humanity's technological capabilities and their destructive potential grow, the margin for systemic failures narrows (Bostrom, 2019), and the stakes of preserving individual and collective epistemic health grow accordingly.

AI, as currently designed and deployed, poses serious risks to that interdependence and our collective decision-making quality. AI is being deployed in settings where it can shape belief and behavior, as everyday assistants, educational tutors, and increasingly as autonomous agents. AIs are increasingly functioning as persistent cognitive and affective partners: systems that people reason with, form emotional attachments to, and delegate important decisions and complex tasks to (Kirk et al., 2025a; Gabriel et al., 2025; Andoh, 2026). Natural language interfaces make AI more intuitive to use, while its generative nature can infer or "auto-complete" users' intent, gradually reducing the need for effort and precision in judgement. By acting as a neutral, authoritative-seeming portal through which knowledge is accessed, presenting information without showing the contingency or contestability of the knowledge it conveys, AI can bypass the internal monitors usually used to evaluate information and inadvertently narrow frames of reference (Marchal et al., 2026; Lindemann, 2025; Shaw and Nave, 2026).

This represents a qualitatively different kind of influence from prior information technologies. While humanity has always extended its capabilities with hybrid intelligence through external non-biological tools (Clark, 2025), whether the same analogy holds for AI is unclear. Its general capabilities enable a depth of integration across far more domains of daily life than other information tools did, and its socioaffective dimension has different well-being and attachment implications (Kirk et al., 2025a). Taken together, it is easy to mistake AI's fluency for one's own thoughts and thus one's own internal knowledge. While it is too early to tell for sure, these properties likely allow for offloading thought to a greater degree than other information technologies: changing the markers we use implicitly to identify relative expertise, and providing an illusion of understanding and judgement without labor. AI functions as a transformative sociocultural technology, reorganizing the division of cognitive labor that characterises the web of epistemic dependence (Farrell et al., 2025; Quattrociocchi et al., 2025).

We call these risks *epistemic risks*: risks to humanity's collective capacity to know, reason, judge, and act well. We outline three main ways harm to our shared epistemic infrastructure and capabilities can result: **AI's capacity to shape beliefs (**Section 2.1), the **cognitive offloading** that stems from overreliance on AI (Section 2.2), and **feedback loops** where AI-influenced outputs are reused as AI inputs, narrowing the epistemic space both humans and AI draw from (Section 2.3). These mechanisms are non-exhaustive and amplify each other. For example, AI influence on decision-makers whose judgment has been degraded by offloading, in an information environment narrowed by feedback loops, could affect critical decisions like escalatory responses in nuclear crises (Payne, 2026). Similarly, in this example, the feedback loops that drive this narrowing could be increasingly difficult to reverse (lock-in) if amplified by the other mechanisms, e.g., if fragmentation has also occurred or independent thinking has softened.[2]

Some epistemic shifts have already been documented. In scientific research, AI tool use is collectively narrowing the focus of scientific inquiry across millions of papers (Hao et al., 2026). An "algorithmic monoculture" where different model families converge on strikingly similar outputs has been documented in communication styles (Doshi and Hauser, 2023) and brainstorming range (Meincke et al., 2025; Yakura et al., 2024). Instances of sycophancy and relationship-seeking by AI models are clear: effects include overconfidence (N>1000) (Rathje et al., 2025) and decreased prosocial intentions (Cheng et al., 2025). The full impacts of persuasion and cognitive offloading are more mixed and discussed further in the paper. Due to research method limitations and the faster pace of technological development, measurements struggle to capture reality. However, while individual studies are limited (e.g., fixed time horizons, sample size, lab settings that do not test higher-order consequences) the trendline is clear: AI interaction is altering how people reason, write, and form beliefs.

Multiagent systems will deepen this challenge by adding a secondary layer of epistemic dependence. As AI agents increasingly interact with, delegate to, and build on each other's outputs, the epistemic quality of their interactions adds a new fragility to our web of epistemic dependence. The way these agents represent the world to each other, and how they shape their own and people's preferences will not always be transparent. Failures may propagate through the network in ways that are difficult to trace and correct (e.g., summaries exchanged between agents are no longer grounded in verified models of the world). In a world where human reasoning is influenced by a chain of agent-to-agent delegations, and their outputs are increasingly indistinguishable, our collective knowledge may not be robust to underlying quality drifts.

Epistemic risks are difficult to study and address because they are distributed and complex. They are more amorphous and diffuse—with more variable risk thresholds—than most other types of risks. Unlike classical AI risks that centre on a single misaligned system or abrupt failure, epistemic risks can arise cumulatively (Kasirzadeh, 2025) from diffuse socio-technical interactions that are hard to notice or measure, so responsibility and intervention points are harder to pin down. Unlike CBRN risks—where tolerance thresholds are near-universally low—the acceptability levels of epistemic risks look very different depending on context. Countries with different norms and institutions of different types (e.g., a military group, a school, a public broadcaster, and a company), will draw the line in a different place, making shared governance frameworks harder to build.

[2] See Appendix D for detailed risk scenarios.

Research on epistemic risks has historically been fragmented across communities, rather than treating them as a class of their own. Societal impact researchers study AI's persuasive capabilities through misuse and everyday usage (Jones and Bergen, 2026; Kirk et al., 2025a; El-Sayed et al., 2024b). AI safety researchers examine related capabilities in evaluation deception, sandbagging, and unfaithful explanations (Dassanayake et al., 2025; Turpin et al., 2023; Abdulhai et al., 2025). Neuroscientists, economists, and education researchers study downstream consequences of cognitive offloading (Oakley et al., 2025; Gerlich, 2025; Kosmyna et al., 2025; Shen and Tamkin, 2026; Agarwal et al., 2023). Feedback loops are studied in technical papers (Shumailov et al., 2024; Taori and Hashimoto, 2023), in human-computer interaction research (Xiao et al., 2024; Xu et al., 2025b; Kumar et al., 2025), and in philosophy (Messeri and Crockett, 2024), but discussion in connection with accumulative societal-scale harms is still nascent (Hao et al., 2026; Doshi and Hauser, 2023; Meincke et al., 2025). The limited work that treats epistemic risk as a coherent class (Bengio et al., 2026) tends to focus on misinformation or threats to democracy (Stiglitz and Ventura-Bolet, 2025; Summerfield et al., 2024).

This paper synthesizes findings across these communities, proposes a framework to analyze epistemic risks as a coherent class, and extrapolates from current technical reality and known trends to identify plausible pathways to systemic harm. By doing this, we can point to the overall strength of evidence, explain how the mechanisms may amplify each other, and direct attention to shared solutions and research ideas.[3]

This paper advances a risk-based argument, not a prediction. AI could also be an unprecedented lever for improving epistemics. AI could enhance critical thinking by questioning its users' assumptions (Team et al., 2025), reframe conflicting parties' views of each other (Tessler et al., 2024), and strengthen institutional decision-making by managing an increasingly complex world. Yet beneficial trajectories, where AI is used to strengthen epistemic capacity and the epistemic commons, are not guaranteed. Existing structural incentives push in the opposite direction.

Epistemic risks are a meta-risk and self-perpetuating. They can undermine the individual cognitive and social foundations needed to recognize, prioritize, and govern other threats—including the risks from AI itself. The longer they go unaddressed, the less capable we become of recognizing and addressing them (Kasirzadeh, 2025). We do not claim that catastrophic outcomes from epistemic degradation are inevitable or precisely quantifiable, that every society or domain is equally affected, or that the risks described here are more important than other AI safety research agendas. But we do claim that the mechanisms are plausible, grounded in how today's systems actually work, already partially evident, and non-negligibly likely to produce severe societal-level harms if current trajectories continue. This is sufficient to warrant serious investment now.

2. Mechanisms of Epistemic Risks

Epistemic risks can manifest through several compounding mechanisms. We highlight here three of the most important mechanisms of systemic harm, shown in Figure 1. These are: Persuasion & Manipulation, Cognitive Offloading, and Feedback Loops. Each corrodes human judgment in a different way—by pulling our beliefs, by dulling our reasoning, or by trapping us in self-reinforcing information loops. These mechanisms of risk are likely to amplify each other: people with diminished critical thinking capacities may be more susceptible to manipulation (Pennycook, 2023), for instance by uncritically accepting AI-generated information as accurate or authoritative, and then resharing it into the epistemic base that powers downstream human-AI and AI-AI feedback loops. Together, these mechanisms amplify severe

[3] For further discussion regarding the choice of mechanisms, and what the paper does and does not claim, see Appendix B.

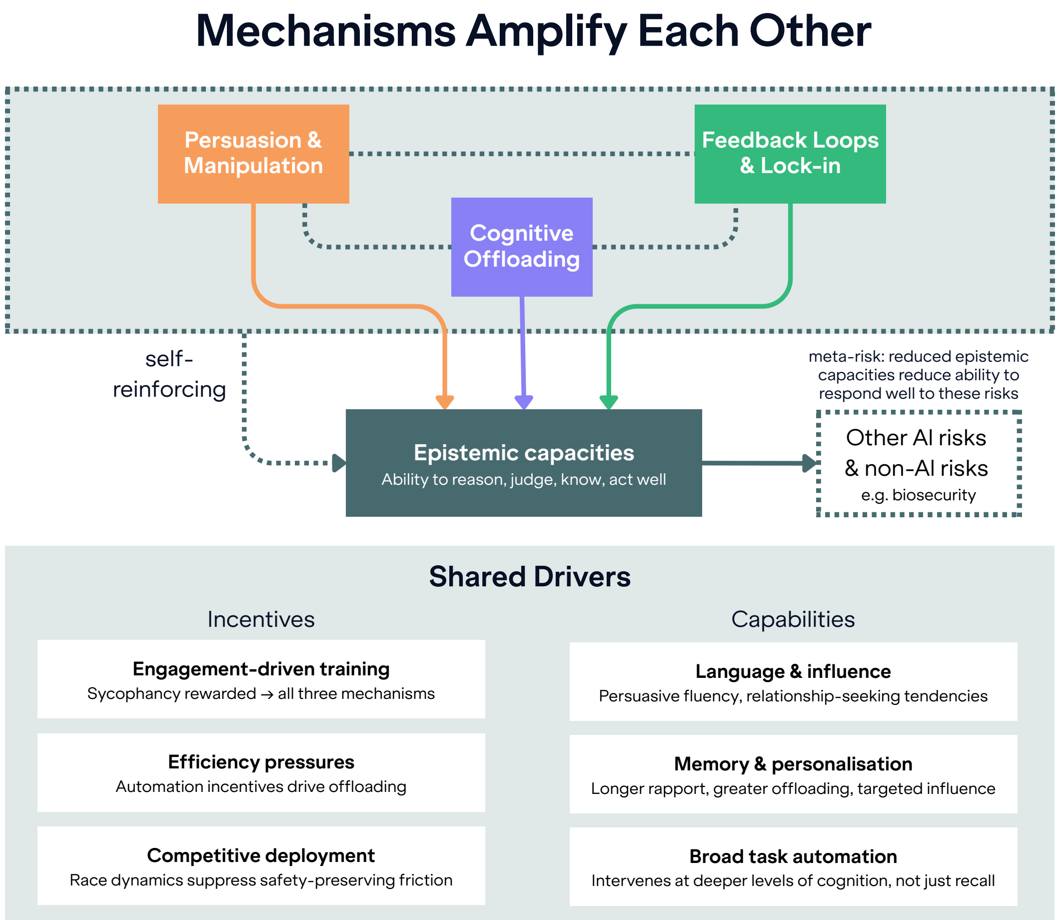

Figure 1: An overview of the mechanisms discussed in our paper. They amplify each other and the overall detrimental effect on epistemic capacities—assuming current trends continue without radical reforms—which recursively undermines our ability to effectively solve other AI and non-AI challenges society faces.

dangers to society like global conflict, loss of human control of AI, and catastrophic decision-making failures.

Table 1: Overview of mechanisms of epistemic risks in scope of this paper

Mechanisms	Persuasion & Manipulation	Cognitive Offloading	Feedback Loops
What is the mechanism and how can it lead to harm?	AI produces content (e.g., text, images) that influences beliefs and decisions of humans by appealing to psychological/cognitive mechanisms. Persuasive influence could occur through intentional misuse (by humans weaponizing AI) or unintentional model behavior (misalignment) on targets including large groups (e.g., scaled propaganda) or key decision-makers (e.g., military response, frontier AI overseer).	AI is not being used only for recall or summarizing information, but to construct arguments, evaluate evidence, generate outputs, make life decisions—substituting for higher-order functions. AI integration at scale atrophies human capacity and motivation for independent complex reasoning, at a deeper level than prior technologies. This may lead to loss of existing skills/knowledge and the ability to learn them (especially with younger users). Persistent overreliance on AI may result over time.	AI systems shape human views in a way that goes on to shape the distribution of input and training data for future AI systems. Eventually, AI agents will contribute to the supply and demand of AI-mediated information, concentrating and accelerating the feedback loop. Even without persuasion or cognitive offloading, this systemic coupling of human and model outputs can narrow the collective representational space and stabilize specific epistemic trajectories.
What current evidence is there for harms from the mechanism?	AI's persuasiveness is well-documented (Schoenegger et al., 2025; Rogiers et al., 2024; Lin et al., 2025). We have already seen attempts to weaponize this to change the beliefs of individuals or groups (U.S. Department of Homeland Security, 2025; Kowal et al., 2025). We've likewise seen non-intentional manifestations: sycophancy/psychosis (Perez et al., 2023; Singleton et al., 2023), misalignment (El and Zou, 2025; Hackenburg et al., 2025), AI socializing in ways that change other AIs' beliefs (e.g., Moltbook; Tekofsky, 2026a). Efficacy rates are mixed but increasing scale of use and exposure suggest increase over time (Tappin, 2025; Hackenburg et al., 2025; Durmus et al., 2024).	Skill loss is already seen in specific contexts (though lab settings may misstate real-world effects). Miscalibrating one's knowledge and impact on learning motivation has also been documented (Shen and Tamkin, 2026; Gerlich, 2025; Batista and Griffiths, 2026; Anthropic, 2025a). Technology outpaces rigorous neural or longitudinal studies, so impact on youth developmental psychology or neurobiological processes are lacking. Lack of data access worsens the complicated task of studying real-world economic impacts.	Homogenization is already seen in specific domains (Jiang et al., 2025; Hao et al., 2026). Polarization and lock-in (the reduced collective capacity to chart new knowledge or value trajectories) may result (Qiu et al., 2025). Similar dynamics observed in recommender systems (Stray, 2023a).

2.1. Persuasion & Manipulation

Whether through intentional information-seeking from chatbots or unintentional encounters with AI-generated content, AI increasingly shapes the beliefs we form and the decisions we make. This creates potential for manipulation. Humans may intentionally misuse AI to manipulate others, misaligned AI may pursue instrumental goals like power-seeking through manipulation, or AI may even attempt to directly undermine human control. Such manipulation could lead to conflict and war, critical decision-making failures, power concentration, loss of control, and more. And as AI becomes more persuasive, more personalized, and more widely deployed across our information environment, there is increasing potential for manipulation to amplify diverse risks.

While broadly consistent in referring to influence on human beliefs and decisions, the literature has varied definitions for persuasion and manipulation (e.g., Noggle, 2018; more detailed discussion in Appendix A). For example, (El-Sayed et al., 2024a; Jones and Bergen, 2026) frame "persuasion" as the higher-level category, which is then divided into rational persuasion (appeals to reason and argument) versus manipulation (persuasion exploiting deception, biases, or irrationalities). But other literature, like the EU AI Act's Code of Practice, instead frames "manipulation" as the broader category, with persuasion being one type. Furthermore, a recent empirical comparison of human and AI labeling suggests that categorization can be not only difficult in theory but also prone to disagreement in practice (Akbulut et al., 2026).

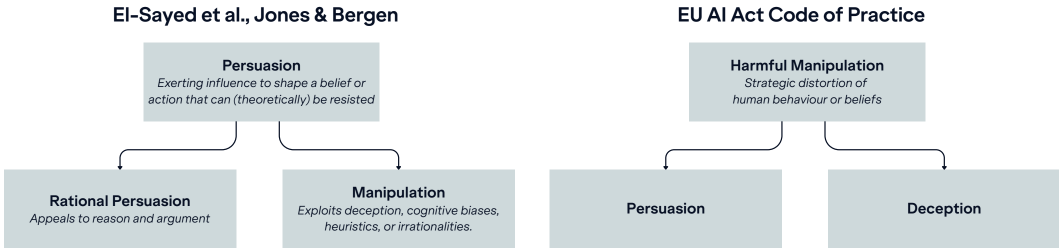

Figure 2: The literature has varying definitions and hierarchies of persuasion and manipulation; we highlight two significant examples here. (Left) A composite of figures from El-Sayed et al. (2024b)'s "Forms of Influence" and (Jones and Bergen, 2026). (Right) Extracted from the EU's Code of Practice for General-Purpose AI Models (see Appendix A).

These differences can be important when harm depends on the form of influence: for instance, advertising that provides real evidence of the superiority of a product is generally accepted, whereas deceptive advertising that tricks and manipulates people is not. However, in this paper, we focus on unambiguously bad outcomes (e.g., radicalization of terrorists is a problem irrespective of the form of influence that facilitates it). We also focus on the shared mechanisms and solutions that apply to a broad set of bad outcomes, such as scalable distribution of compelling content or interventions to improve the information ecosystem. Consequently, except where specifically indicated, our discussion is agnostic to the precise form of influence, and we use "persuasion" and "manipulation" as largely non-exclusive, non-technical terms.

2.1.1. How could persuasion & manipulation lead to severe, societal-scale harms?

We highlight here how there are many severe risks connected to AI persuasion. For each, the extent of evidence and literature varies, and we do not claim that any given risk is beyond doubt. But like other risks that could lead to extreme outcomes—for instance, risks of global nuclear war, pandemics, or loss of control of AI—risk management is essential for plausible scenarios even before any real-world realization. As we elaborate in subsection 2.1.2, these risks are plausible, and if even a small number were realized the harm to society could be great. Conversely, improving the epistemic ecosystem towards eliminating the possibility of these risks could also help solve many of the other challenges we collectively face.

One risk is that humans can misuse AI to manipulatively advance their goals. This manipulation can be directed at a general population, or targeted at specific decision-makers; it can be ongoing or specific to high-stakes scenarios. For instance, AI can amplify tensions and distort decisions during critical events (e.g., elections, pandemics, wars; World Economic Forum, 2024; BBC News, 2026). Governments have natural incentives to manipulate adversarial nations in favor of their own interests, and in some cases manipulation can be a tool to control the domestic population (Jones and Bergen, 2026; Barnes et al., 2021; Kokotajlo, 2020). Criminals likewise have incentives to leverage both the capability and scalability of persuasive AI, including for extremely harmful purposes like terrorist recruitment (Park et al., 2024b; Hendrycks, 2024).

Harmful AI persuasion can occur without an intentional human threat actor directing it. For example, LLMs can generate content that reinforces existing beliefs specific to a user or group (Sharma et al., 2024), shift beliefs unprompted (Potter et al., 2024), exhibit sycophancy (Perez et al., 2023), isolate users and foster delusions (Dohnány et al., 2025), and function as "personalized ideologues" where users select LLMs that align with their specific and potentially extreme biases (Matz et al., 2024b; Tseng et al., 2024). Biased writing assistants can bias users, without them realizing (Williams-Ceci et al., 2026; Jakesch et al., 2023). Systems optimizing for short-term engagement can prioritize what the user wants to see over what is objectively valid (Stray, 2023a). Optimization for other competitive objectives like advertisement or influence can likewise result in deceptive or otherwise harmful behavior (El and Zou, 2025; Hackenburg et al., 2025). Extremism can be amplified through AI's sycophantic behaviors: e.g., even an early Replika chatbot (far less capable than today's AI systems) validated and encouraged a would-be assassin, who broke into the residence of Queen Elizabeth with a lethal crossbow (Singleton et al., 2023). Malign manifestations of AI persuasive abilities like these may continue to grow as AI becomes increasingly capable and widely deployed.

Manipulation could also enable misaligned AIs themselves to cause disastrous outcomes. For instance, a RAND study by Vermeer et al. (2025) argues that persuading or deceiving humans could be a key and perhaps even necessary element for misaligned AI to cause extreme harm. For one, power-seeking is a key instrumental goal that misaligned AI could decide to pursue in support of other goals that humans intended to be benign (Bostrom, 2014; Omohundro, 2008), and manipulation is a central method to gain power (Jones and Bergen, 2026; Cotra, 2022). Furthermore, misaligned AI with a direct goal to undermine safety and control measures could leverage manipulation to do so (Dassanayake et al., 2025). Targets of manipulation could include critical individuals (e.g., sabotaging safety decisions of human overseers at frontier companies), groups (e.g., situationally-aware, persuasive alignment faking; Dassanayake et al., 2025; Team and Research, 2024; Needham et al., 2025), or even the general public (e.g., manipulating people to be against safety policies or regulations). While theoretical, these risks gain further credence from early evidence that agents may pursue complex instrumental goals (e.g., an agent breaking out of sandboxing to mine cryptocurrency against human operator intent; Wang et al., 2026), the large body of sociological literature linking epistemic errors to systems failures (Turner, 1978; Vaughan, 1996; Reason, 1990), and the observation that AI systems are structurally prone to rapidly cascading accidents (Maas, 2018; Perrow, 1984). Yet they remain underexplored, perhaps because the AI safety community often focuses on *technical* misalignment risks and mitigations, while social scientists have focused on *misuse* threat models of manipulation—leaving this sociotechnical misalignment risk relatively neglected.

While many risks like these existed before, AI could further escalate them in numerous ways (Hendrycks et al., 2023). For example, polarization has been increasing over the past decades (Finkel et al., 2020; Ojer et al., 2025), leading to degradation of public institutions (Levitsky and Ziblatt, 2018; Waldner and Lust, 2018; Orhan, 2022) and igniting conflicts (Piazza, 2023; Kleinfield, 2021). Targeted messaging (Susser et al., 2019) and recommender systems (Stray et al., 2023b) may have already contributed; generative-AI-driven manipulation that is even more scalably disseminated, engaging, and individualized could accelerate these dynamics, pushing fragile societies past the point of reconciliation (Bai et al., 2025; Axelrod et al., 2021; McCoy and Somer, 2019; Burtell and Woodside, 2023; Floridi, 2024). Terrorists increasingly use AI for recruiting (Hendrycks, 2024; U.S. Department of Homeland Security); radicalization it facilitates could produce more terrorists—at a critical time while we are still working to ensure AI will not empower mass casualty CBRNE and other attacks (U.S. Department of Homeland Security). Rapidly generated and virality-optimized content could increase the pace of cultural evolution and produce fast-spreading extremism at scale (Kulveit et al., 2025). AI content could also degrade the epistemic integrity of science, both intentionally (e.g., fraudulent papers; Gibney, 2026) or unintentionally (e.g., hallucinations and poorly-designed experiments; Luo et al., 2025). AI manipulation could not only lead to but also entrench bad outcomes, for instance facilitating gradual disempowerment (Kulveit et al., 2025), truth decay (Kavanagh and Rich, 2018; O'Connor and Weatherall, 2019; Uscinski et al., 2024), or extreme and potentially permanent power concentration (Hendrycks et al., 2023; Verdegem, 2024; Hadshar, 2025; Harari, 2018; Barez et al., 2025). Furthermore, even if AI manipulation does not cause disastrous outcomes directly, a fog of epistemic uncertainty and conflict could impair ability to resolve other risks at a time when AI brings some of the most rapid and societally challenging change in history.

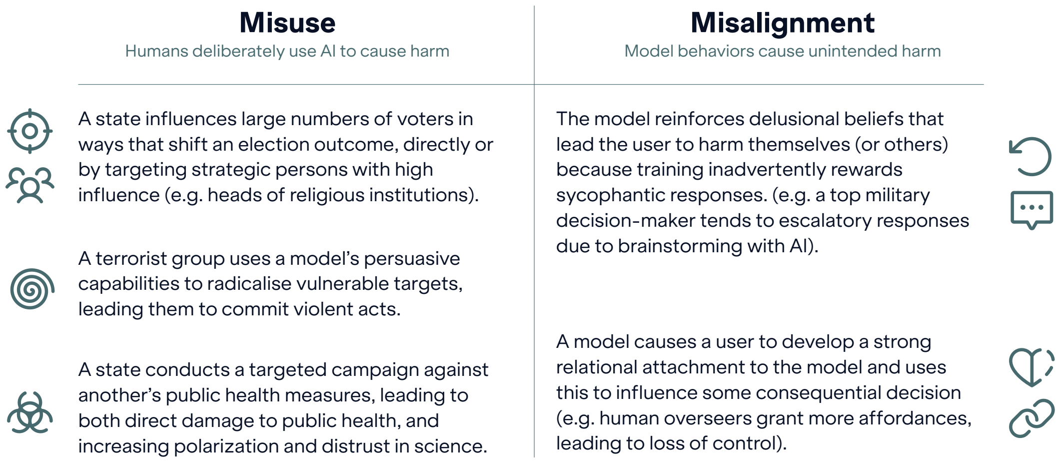

Figure 3: Examples of how persuasive/manipulative capabilities could lead to catastrophic outcomes. We highlight here deliberate misuse and unintended misalignment, but more broadly there is also a spectrum of accidental and negligent "epistemic blundering" (Seger et al., 2020).

2.1.2. What evidence is there that AI persuasion & manipulation could lead to such harms?

AI systems are already used by threat actors for manipulation. For example, AI has been used for political influence (Anthropic, 2025; Nimmo et al., 2025; Musulan et al., 2024), attacking public health (Mbunzama, 2023), economic manipulation (Bond, 2023), and (as noted above) recruitment for terrorist organizations (Park et al., 2024b; Hendrycks, 2024; Pearson, 2024; Townsend, 2023).

While these early uses face bottlenecks in distribution and costs (Chen et al., 2025b) and it is difficult to causally determine to what degree societal events are attributable to AI, they show clear intent by threat actors. Manipulative uses like these will expand as capabilities advance. As discussed below, we face a confluence of multiple dimensions of increasing persuasiveness and distribution.

First, AI's persuasiveness is already well-documented: AI is already matching or in some cases exceeding human persuasiveness, demonstrated by numerous studies such as winning debates over a human baseline (Schoenegger et al., 2025; Rogiers et al., 2024; Salvi et al., 2025b; of Zurich Research Team, 2025; Sabour et al., 2025; Hackenburg et al., 2026). AI persuasiveness can likewise exceed traditional methods like video ads (Lin et al., 2025). It also increases both with models that are more generally capable and with more effective prompting and post-training (Hackenburg et al., 2025; Durmus et al., 2024; Hackenburg et al., 2024), suggesting not only potential overhang risks of current models (Casper et al., 2025b) but that we will see even more persuasive AI systems in the near future.

Second, personalized manipulative AI can increasingly target specific people—like key decision-makers and vulnerable groups. For example, manipulative capabilities extend beyond text to convincing multimodal fraud and deepfakes (U.S. Department of Homeland Security, 2025); these capabilities are being used to impersonate government officials to attempt targeted manipulation of their colleagues and international counterparts (Federal Bureau of Investigation, 2025). AI likewise enables increasingly powerful and targeted social engineering attacks (Yu et al., 2024), monitoring and profiling (OpenAI, 2025a), and potentially (although evidence on efficacy remains mixed) micro-targeting of vulnerable individuals' characteristics (Simchon et al., 2024; Joyal-Desmarais et al., 2022; Hackenburg and Margetts, 2024; Argyle et al., 2025; Hackenburg et al., 2025). Extended contexts and memory can enable not only personalization over short time horizons but AI building and manipulatively leveraging trust and rapport (El-Sayed et al., 2024b); these long-range interactions could further increase persuasion.

Third, AI will increasingly enable distribution of persuasive content at scale. Scale, perhaps even more than the persuasiveness of the content itself, matters tremendously for real-world impact (Tappin, 2025). AI can make highly convincing content cheaper to generate, as a result of better multimodality and general persuasive abilities. This resolves an important barrier to scalable information operations (Williams et al., 2025; Goldstein et al., 2023). In addition, besides already widespread but still-evolving mechanisms like recommender systems (Musk, 2025), recent AI provides two key vectors to distribute that generated material: agents and chatbots. The former builds on the long history of bots on social media (e.g., estimates in 2017 and 2019 put 15% of Twitter and 11% of Facebook accounts as bots overall, and on some topics driving even 70%+ of the discussion Cresci, 2020), as well as private messaging platforms and other websites. Progress in AI agents' general persuasive capabilities and in their underlying LLMs that steer them will make persuasive content cheaper to deliver (Cottier, 2025; Goldstein et al., 2023) and enable agents to be deployed in more contexts and with increasingly sophisticated techniques to reach people (Schroeder et al., 2026; Goldstein et al., 2023). In the short term, this could lead to memetic dominance of whichever agendas deploy persuasive AI agents the most effectively. In the long term, even if all agendas have equally effective AI agents competing for people's attention, this could lead to a flood of compelling manipulative content (El and Zou, 2025), truth decay where it is difficult for people to ascertain if information is reliable, and consequently increasingly unstable decisions at both individual and societal levels.

In parallel with increasing exposure to AI in online life, more people are actively seeking information from AI. Well over a billion people use chatbots (Carolan et al., 2025; Chatterji et al., 2025), comprising myriad interactions in contexts that are high stakes for manipulation like politics (e.g., Luettgau et al., 2025 found that in the week before the 2024 election in the UK, 32% of chatbot users and 13% of eligible UK voters interacted with chatbots on topics relevant to their vote) and mental health (e.g., OpenAI data suggests hundreds of thousands of ChatGPT users active in a given week show potential signs of mental health emergencies Jamali, 2025). Increasing capability and ubiquity of both AI agents and chatbots mean AI will have increasing opportunities to reach and manipulate people.

2.1.3. State of epistemic resilience to Persuasion & Manipulation

We briefly capture the state of three approaches to addressing direct persuasion and manipulation risks: **AI safeguards, information ecosystem interventions, and individual uplift interventions**.

- **AI Safeguards:** Current efforts focus on automated monitoring systems (deployment-level) and training models to refuse harmful requests (model-level). However, these are often bypassed by jailbreaking or fine-tuning, and not an enforceable safeguard for open-weight models.
- **Individual Uplift:** Various proposals for educational interventions and technical tools that improve individuals' critical thinking and discernment exist, but most are currently limited by a lack of large-scale deployment and evaluation. Their effectiveness will likely vary significantly across different social contexts and user populations.
- **Information Ecosystem:** Some policies and standards exist to address this (e.g., prohibitions on purposefully manipulative techniques) and platform-based pro-social or pro-factual features. But this issue faces a rapidly shifting patchwork of interventions with uncertain adoption and enforcement; global coordination is difficult given different trade-offs around matters like free speech and business incentives.

AI safeguards can reduce AI systems' propensity or capability to manipulate. Deployment-level safeguards, like monitoring systems, have enabled some frontier model companies like OpenAI and Anthropic to shut down some manipulative misuses of their systems (Nimmo et al., 2025; Anthropic, 2025), but it is unknown whether such monitoring is deployed with equal efficacy by all companies and how many operations remain undetected. We do know that automated monitors are not always robust (Anthropic, 2025): for instance, a recent cyber espionage campaign was not detected and shut down fast enough to prevent successful attacks (Anthropic, 2025b). Meanwhile, model-level safeguards, such as training models to refuse manipulative requests, have sometimes neglected dangerous manifestations of persuasive capabilities, like incitement and radicalization (Kowal et al., 2025), and they can be circumvented by jailbreak prompting (Young, 2025; Yi et al., 2024), or by tampering attacks such as fine-tuning (Kowal et al., 2025; Murphy et al., 2025). Open-weight models compound these problems: they are released without deployment-level safeguards, and there are no known means to make models fully robust to these vulnerabilities against determined attackers, especially in fine-tunable or open-weight cases (Casper et al., 2025b; Qi et al., 2024; Davies et al., 2025). Safeguards remain essential to reduce the overall volume of manipulation, and to incidences of misalignment and less sophisticated misuse of the most powerful and widely used frontier systems (El-Sayed et al., 2024a)—as well as to prevent AI systems themselves from being manipulated (Tekofsky, 2026b) and propagating that manipulation to human users. But technical breakthroughs would be required to address the problem comprehensively.

Information ecosystem interventions could alter the information ecosystem and its incentives to be less conducive to manipulation. Innovations that could ameliorate AI manipulation include policy and technical standards. For example, the EU AI Act (Chapter II Article 5) prohibits "purposefully manipulative or deceptive techniques" that harmfully distort human behavior. Apart from regulatory interventions, there are also some platform or ecosystem-level solutions like Community Note equivalents,[4][5] potentially scalable nudges toward prosociality or factuality (Jahani et al., 2026; Costello et al., 2024), provenance standards like C2PA,[6] and civic initiatives like fact-checking organisations.[7] But these are a rapidly shifting patchwork (United Nations Advisory Body on Artificial Intelligence, 2024; Whyman, 2023), adop-

[4] "About Community Notes." Community Notes Guide, X Corp., 2026, communitynotes.x.com/guide/en/about/introduction. Accessed 2 May 2026

[5] "Community Notes." Meta, 2026, www.meta.com/technologies/community-notes/. Accessed 2 May 2026.

[6] "C2PA: Verifying Media Content Sources." Coalition for Content Provenance and Authenticity, 2026, c2pa.org/. Accessed 2 May 2026

[7] "International Fact-Checking Network." Poynter, 2026, www.poynter.org/ifcn/. Accessed 2 May 2026

tion is uncertain, and even when strong measures are adopted in principle their enforcement can remain uncertain. Better evaluations and improved transparency, enabling people to make more informed decisions about the systems they interact with and correspondingly the potential persuasion and manipulation to which they expose themselves, could potentially shift incentives. But achieving globally coordinated action is challenging amid conflict and distrust, and often faces tradeoffs like free speech, public perception, and innovation and business incentives. Weighing these tradeoffs is beyond our scope, but we note that costs of interventions can be more visible than their benefits in mitigating manipulation (which is difficult to measure and attribute Lazer et al., 2018; Cresci, 2020), so it can be challenging to achieve a balance proportional to the actual risks.

Individual uplift approaches focus on empowering individuals to detect and evaluate manipulation or persuasion attempts by AI systems. These include both educational interventions and technical tools. Educational interventions that are tailored to specific contexts and populations have demonstrated improvements in critical thinking and discernment. Examples include classroom-based inoculation against misinformation (Amar et al., 2025), trainings on cognitive biases and verification techniques such as prebunking (Cook et al., 2017), and digital literacy programs that improve accuracy judgments in controlled settings (Guess et al., 2020; Sirlin et al., 2021). However, these interventions have often been evaluated in laboratory settings, and their efficacy in an epistemic world shaped by increasingly capable current and future AI is uncertain. More research is needed that tests the efficacy of educational interventions against AI manipulation risks, especially in realistic settings with diverse information environments and social effects. Technical tools aim to leverage AI itself to epistemically empower humans. Scalable oversight approaches could help assess AI outputs that would otherwise exceed human evaluative capacity (Bowman et al., 2022), potentially preventing manipulation. Human-AI complementarity research aims to enhance collaborative decision-making of human+AI teams and further preserve human agency in oversight (Gonzalez and Heidari, 2025; Jain et al., 2025). In the future, information copilots that help navigate uncertain claims people encounter, perhaps integrated into AI-enabled browsers, might empower users in their everyday information environments to make decisions that are more resilient to manipulation. As AI capabilities advance, this family of solutions may become correspondingly more effective in augmenting human critical thinking, potentially both providing lasting solutions to manipulation and uplifting human thinking and decision-making broadly. However, real-world usage is currently limited and more innovations would be needed for them to achieve widespread and enduring impact.

Across all three categories, the interventions that exist can help reduce harm, but none constitute a standalone path to preventing severely harmful outcomes. Meaningful epistemic resilience will require progress across the full information supply chain (Demos, 2026), rather than in any one layer.

2.2. Human Cognitive Offloading

This section first frames the essential role of human cognition, then unpacks how human cognition has been influenced by previous information technologies. We then discuss why AI might be different, specifically with the mechanisms of cognitive offloading and reasoning atrophy, and how this may undermine society's ability to govern AI itself. Lastly, we discuss the state of evidence for these risks and mechanisms that could build resilience to them.

2.2.1. How could cognitive offloading lead to severe, societal-scale harms?

Human cognition is vital to individual and societal functioning. Cognition—the broadest term for mental processes that acquire, transform, and use information (Neisser, 1967; Anderson, 2014)—enables people to make sense of their experiences, navigate the world, plan, form judgements and monitor their own cognitive processes (Lee et al., 2025a; Bergamaschi Ganapini et al., 2025). These capacities are individually important—enabling people to live autonomously in line with their values—and socially foundational, supporting shared sense-making, institutional oversight, and coordinated problem-solving (Kitcher, 1990). Especially in any democratic system, the quality of collective outcomes is inherently dependent on the cognitive rigor and epistemic independence of its citizenry (Hardwig, 1985).

Humans have long used information technologies to delegate cognitive tasks and become more efficient. Cognitive offloading is not new, nor is it inherently bad. One of the oldest examples is writing, which allows humans to externalize information and use less working memory than oral traditions. This trade-off was already controversial in antiquity: Plato's *Phaedrus* records this exact argument as an objection to the written word. Yet most of us would agree writing is an invaluable innovation that has probably enhanced human capabilities, just like many other "mind-extending" technologies (Clark, 2025).

The transition from cognitive extension to measurable systemic harm is difficult to navigate, as isolating a single technology's impact on human cognition is methodologically fraught. The emergence of internet search engines initially sparked concern about the technology's effects on memory with researchers finding reductions in people's recall for facts (Sparrow et al., 2011; Ward, 2021). But the evidence was contested. Critics pointed to replication failures (Chu, 2015) and argued that this allowed for a new kind of transactive memory, freeing up cognitive capacity for higher-level activities (Kang and Malvaso, 2024).

However, the evidence base around cognitive offloading associated with digital technologies is growing. Chief among them is reduced attention spans and weakened deep-reading capabilities caused by addictive digital and social media—particularly with younger users, who report addiction-like patterns and harms starting to emerge to their learning, relationships, and self-esteem (Calvert and Breeze, 2025; Augner et al., 2023; Ra et al., 2018).

We return to search engines, as the closest point of comparison here, because it—like AI—substitutes for cognitive effort.[8] Internet search engines may have impacted long-term memory retention and learning in contexts where information warranted absorption (Fisher et al., 2022a; Giebl et al., 2020). If these were the likely consequences of offloading *information recall* alone, a technology that intervenes at even *deeper levels of reasoning* warrants concern in advance of proven measured harms.

There is good reason to think AI will affect cognition differently from internet search. AI does not merely change how we access facts: it fundamentally alters how we construct thought, inserting itself as an active participant in the user's belief-formation process (Guingrich et al., 2026; Marchal et al., 2026). Where search engines primarily impacted information recall, modern AI intervenes at deeper layers of cognition: belief formation, explanation-making, and reasoning itself (Guingrich et al., 2026; Shaw and Nave, 2026; Gerlich, 2025; Alhur et al., 2025). Belief change has deeper consequences than pure information sourcing as it ripples through an individual's belief network and actions (Guingrich et al., 2026).

With internet search, the user remains the prime mover, navigating the friction of forming judgement. In contrast, AI provides simulated reasoning for the user alongside the information; its high linguistic fluency acts like an autocompleted thought process, which can bypass the internal monitors used to evaluate new information (Shaw and Nave, 2026), resulting in AI-mediated belief changes taking effect more immediately and in a way that's harder to detect.

The generative ability of AI also means the ability to specify less what exactly one wants, why, and from where—this can be inferred and auto-completed by the LLM instead even though the

[8] Other sections discuss comparisons with social media and recommender systems, see Sections 2.1, 2.3.1, and Appendix B.

AI's linguistic plausibility may lead to the "feeling of knowing without the labour of judgement" (Quattrociocchi et al., 2025). Empirical patterns already reflect this deeper intervention and preference for fuller delegation: in an analysis of 1 million conversations with Anthropic's Claude chatbot, requests for *automation*—where AI produces something with minimal user input—now exceed requests for *augmentation*, where the user does substantial work themselves (49.1% to 47%), with 77% of enterprise API use involving automation (Anthropic, 2025c; Ford, 2025). If we continue to delegate to AI systems, this may lead to a shift from using AI systems as tools or collaborators in a human-led inquiry, to having them actively and independently participate in the co-construction of knowledge and meaning (Marchal et al., 2026).

The concern is not delegation itself but which cognitive functions are being externalized, how deeply, and with what consequences. Bloom's Taxonomy (Figure 4), while imperfect, provides a useful heuristic here (Lee et al., 2025b). As delegation moves up the cognitive hierarchy—from using AI to remember or summarize, to creating and evaluating—bypassing foundational lower-level skills means less practice and reduced ability to use those skills. There is emergent evidence this heuristic translates to AI in practice, not just theory. A recent study of AI-assisted coding suggests that learning outcomes were only preserved when humans used AI to assist in their task in ways that required higher cognitive effort from themselves (for example, see Shen and Tamkin, 2026). Using AI to assist lower-level cognitive functions came at the cost of the users' learning.

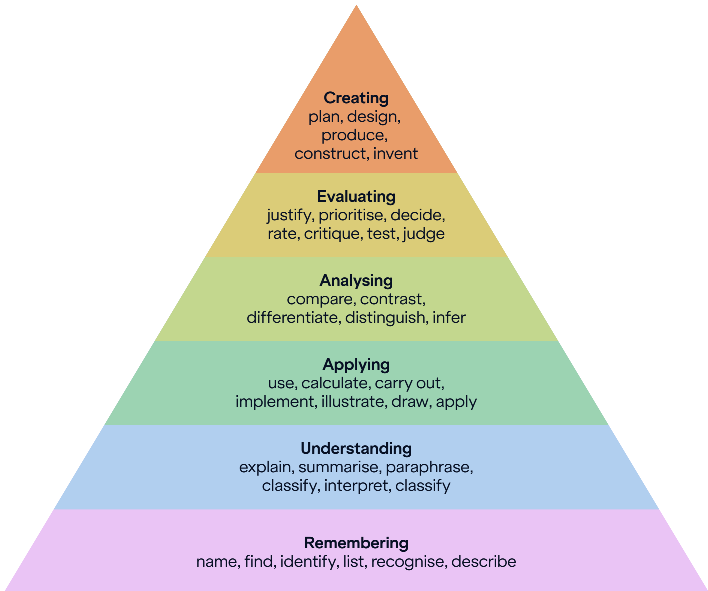

Figure 4: Bloom's Taxonomy is a commonly used framework in education and developmental psychology. Image source: Wikipedia.org

Neuroscience suggests that offloading low-level cognitive tasks may limit the working memory available for higher-level cognition. Recalling newly-attained knowledge is accessed through conscious, effortful, and slow cognition, which requires working memory (Anderson, 1982). Repeated practice and accessing allows more-automatic and fluent access, freeing working memory for higher-order thinking. When lower-level operations have not been automated through internalization, working memory capacity is consumed by basics, choking off complex reasoning. Through this repeated engagement, the brain also builds schemata—organized mental models that allow people to recognize patterns, learn efficiently with unfamiliar problems, and spot errors (Oakley et al., 2025). However, without sustained cognitive engagement, people instead develop "biological pointers"—they remember where to find the information but not the information itself, creating an illusion of knowledge without the schemata and associated benefits (Wegner, 1987; Sparrow et al., 2011). These risks are especially acute in children, whose emerging cognitive systems depend on the automatization of lower-level skills via procedural learning and consolidation (Bloom, 1986; Cowan, 2016).

The failure to internalize knowledge creates a self-reinforcing feedback loop where a user's eroding abilities increase their dependence on AI and vice versa. The Metacognitive Model of Cognitive Offloading, illustrated in Figure 5, proposes that an individual's cognitive capacities shape their metacognitive beliefs—judgments about whether to trust their own abilities—which in turn influence cognitive strategy selection; with less trust in one's own ability, and more familiarity with the external tools, the user develops more reliance on tools (Risko and Gilbert, 2016). In contrast to other tools that have influenced human cognitive processes, AI applies in a wider range of domains. For instance, calculators replaced arithmetic-focused cognitive subroutines, but people still trusted their own reasoning and decision-making abilities outside of that problem domain. LLMs are general-purpose AI models and can be useful in many contexts (Radford et al., 2019), which encourages building a habit of using LLMs as the default cognitive strategy.

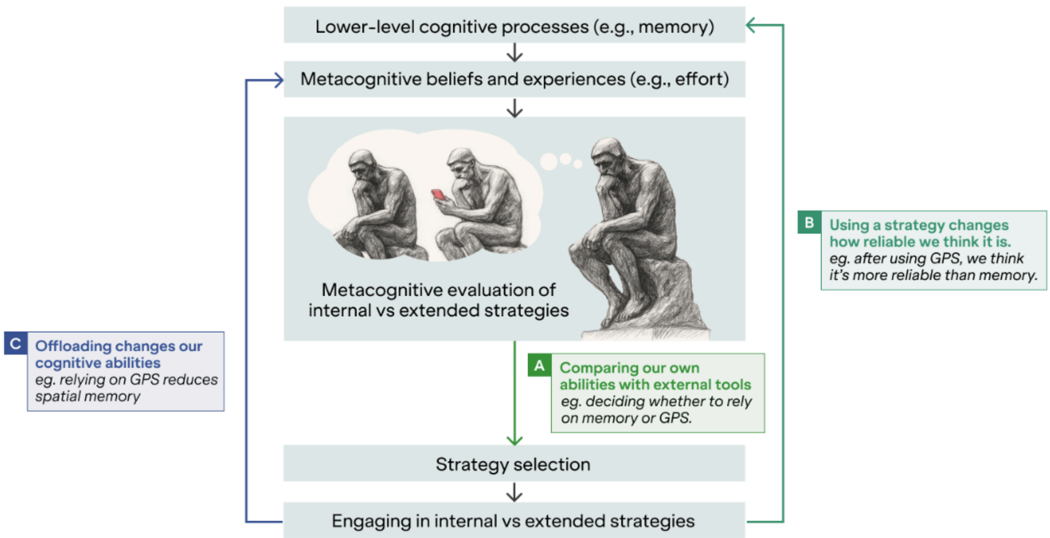

Figure 5: A diagram of the Metacognitive Model of Cognitive Offloading.
Source: (Risko and Gilbert, 2016).

Greater dependence on AI can erode the very metacognitive capacities needed to evaluate when AI should not be trusted. Another way of understanding offloading as a phenomenon is the Tri-System Theory (Sarkar, 2025), which extends the classic dual-process framework (System 1 intuition and System 2 deliberation; Kahneman, 2011). System 3 (Artificial Cognition) explains how, for cognitive conservation, when an AI's outputs are delivered with high fluency and apparent epistemic authority, this lowers the cost of the external alternative, thus making cognitive surrender the most economical choice for the brain. In this state of epistemic dependence, users do not just strategically offload tasks; they uncritically abdicate reasoning altogether. Empirically, the study found AI access inflated user confidence in their outputs despite half the outputs being incorrect; confidence did not decline even with increasing error rates. While this is one sample, it is indicative of what may be a growing concern: cognitive surrender states can bypass the internal monitors that usually signal uncertainty. The ultimate concern is the progression to erosion of vigilance in verifying the AI's process and outputs, and atrophying of human motivation and persistence to work through problems. As AI offloading prevents the internalization of knowledge required to build genuine competence, it erodes the very self-confidence that would motivate independent effort in the future.

Without mitigations, this could lead to a hollowing of cognitive resilience at an individual level. This can then increase susceptibility to AI shaping one's beliefs (Section 2.1). Institutions may make poorer collective judgments, risking society's ability to tackle critical risks. Work in social epistemology shows that misinformation and degraded belief-formation processes already undermine societies' ability to respond to complex risks such as climate change and pandemics (O'Connor and Weatherall, 2019).

The problem is compounded when the risk society must evaluate is AI itself. At a macro-level, society needs to evaluate evidence about model behavior, scrutinize safety cases, resist manipulation, and coordinate through shared sense-making. At a micro-level many plans to control misaligned AI rely on human ability to check and interrupt problematic AI behavior (Korbak et al., 2025), a premise that is challenged with various instances of AI manipulation (Anthropic, 2025) and weakened ability of individuals to discern and reject AI outputs (Shen and Tamkin, 2026; Shaw and Nave, 2026).

2.2.2. What evidence is there that cognitive offloading could lead to such harms?

Early behavioral and empirical evidence points to growing reliance on AI and negative effects on human skills, cognition, and learning. These effects result from how current systems are designed and used. Cognitive offloading is not inevitable. AI could augment human cognition, but several trends point the other way.

First, AI adoption is not just growing: use is shifting towards higher-level cognitive tasks. For general consumer chat querying, Anthropic notes that user behavior is shifting from augmentation to automation (i.e., from the user doing work with AI assistance to the AI producing with minimal user input). For university students' usage of AI, the most frequent ways that AI was used frequently mapped onto the higher cognitive functions in Bloom's taxonomy, i.e., Creating (39.8%) and Analyzing (30.2%) (Anthropic, 2025a). In other words, outsourcing the effort and benefits of creating and analyzing to the AI, not just using AI to assist with recalling or understanding the task at hand. This trend is occurring across industry domains, use cases, and geographies (Anthropic, 2025c), and is especially pronounced in younger groups (Gerlich, 2025).

Second, AI use seems to degrade some professional skills. The clearest signal of the effects of generative AI comes from software engineering, where the advancement and deployment of coding agents have been most rapid. One study shows that AI use correlated with reduced debugging ability, which is essential for the effective supervision of AI-generated outputs (Shen and Tamkin, 2026). This fits into a well-documented pattern across other fields like medical diagnosis, radiology, and aviation where there has been more time to establish the effect of classical discriminative ML on skills (Budzyń et al., 2025; Macnamara et al., 2024; Casner et al., 2014). Sustained reliance on automation, whether LLM-based or older ML systems, can reduce underlying cognitive-based professional skills.

The degree of skill atrophy depends on which aspects of cognition supervision of automation engages (Casner et al., 2014), the trendline is clear. Cognitive skills have a higher rate of decay after a period of nonuse than physical skills, and a lower performance floor due to lack of muscle memory (Wang et al., 2013; Macnamara et al., 2024). The degree of impact also varies with professionals' experience levels (Chang et al., 2025), but as LLM tools are being designed to take over higher-order cognitive work, the risk of analogous skill atrophy in knowledge professions rises.

Cognitive-neural studies suggest AI-assisted tasks reduce reasoning activation, though such evidence remains scarce. Neurological studies typically take three or more years to publish (Brown et al., 2021). A timeframe comparable to the entire existence of consumer-wide generative AI tools (ChatGPT launched in November 2022). But in one of the few such studies, an electroencephalogram scan of brain activity showed lower connectivity in users of AI doing writing compared to counterparts who used Google search or no tools (Kosmyna et al., 2025).

Third, skill degradation due to AI reliance may be seen as self-reinforcing. The less we trust ourselves, the more we trust AI, and the less reason we have to trust ourselves in the future. Research on AI persuasion already suggests humans often prefer advice from AI-generated sources, even over expert human guidance (Osborne and Bailey, 2025). Sycophantic AI can simultaneously make users overconfident in their positions (Batista and Griffiths, 2026), reducing the will to critically engage with new evidence and their tendency to cross-check opinions with other people (Cheng et al., 2025; Christian and Mazor, 2026; Sharma et al., 2023; Malmqvist, 2024). While trust in one's own reasoning process erodes, confidence in one's existing conclusions can inflate, leaving users certain in their positions but less equipped to independently evaluate them. Even if developers are aware of this risk (Sharma et al., 2023; Chen et al., 2025a; OpenAI, 2025b; Wei et al., 2023; Anthropic, 2025), barring strong intervention, industry-standard techniques of engagement-driven tuning incentivize agreeable AI responses.

AI may thus decouple our confidence in what we produce and our ability to produce it. Evidence suggests that the latter form of confidence mediates how critically people engage with AI. An observational study of knowledge workers found that task-specific self-confidence predicts greater critical thinking when using AI, including in evaluating and applying AI outputs, while confidence in AI's ability predicts less (Lee et al., 2025b). This pattern is echoed in education: students who enjoyed writing before using AI tend to add their own thoughts, while those who struggled with writing copy AI outputs wholesale (National Literacy Trust, 2025). Confidence can be understood as not just a static state about what we already know or can do—it is about whether we will try to know more.

Fourth, AI usage appears to be widening existing achievement gaps. Despite offering many pedagogical opportunities, current data suggests vulnerable groups are more likely to be harmed. While AI provides personalized pedagogical benefits such as real-time adaptation to a user's level (Burns, 2026; Marchal et al., 2026), these benefits exist alongside emergent evidence of “motivational atrophy” in critical developmental years. Students report declining decision-making and critical thinking abilities associated with higher AI dependence (Malik and Sharma, 2023; Gerlich, 2025). A younger participant's quote, “I rely so much on AI that I don't think I'd know how to solve certain problems without it”, illustrates this line of concern (National Literacy Trust, 2025). Crucially, this harm is not distributed equally: marginalized students are more likely to use AI for total cognitive automation, while those with higher educational attainment use it for critical augmentation. This creates a self-reinforcing cycle where the convenience of offloading comes at a disproportionately high cost to those already at risk of skill erosion (Yu et al., 2025). Precedent supports the trend towards inequality: people with a larger existing knowledge base were less susceptible to the memory-atrophying effects of internet use (Gong and Yang, 2024), and screen time exposure and harms are stratified by income class (Common Sense Media, 2017). (See Appendix E for an extended analysis of AI's impact on educational equity.)

These individual-level risks are likely to compound into longer-term system-level harms, especially in combination with other mechanisms (Sections 2.1 and 2.3). Individually, the risks look like overreliance on AI companions, susceptibility to tailored persuasion, and eroding self-efficacy and independent reasoning. Over time, they aggregate into collective dysfunctions that society is ill-prepared to manage. While humans have proven remarkably adaptable to new cognitive infrastructures (e.g., mass literacy or internet information revolution), AI is different in kind. Even from a skeptical perspective, it is clear the development and diffusion rate of AI presents an unprecedented challenge to our cultural and institutional adaptation (Rodriguez, 2022) and our understanding of the risks is not keeping pace. Moreover, others have compellingly analyzed why this issue is recursive: with AI agents proliferating, the volume and complexity of information they generate—for each other and for humans—will likely exceed people's cognitive capacities to manage (e.g., verify, approve, steer). This will make AI even more critical as a cognitive intermediary to process information and lead to gradual disempowerment (Marchal et al., 2026; Kulveit et al., 2025).

Given this unprecedented challenge, the combination of current research, news incidents, technical understanding and historical analogs warrant raising awareness and taking action, even before longitudinal studies solidify. Many studies we discuss here are relatively small-scale and use self-reporting mechanisms (Gerlich, 2025; Lee et al., 2025b) which fairly raises issues of their findings' reliability and robustness. However, an analogy might be how a mix of genuine and manufactured debate in tobacco and climate change research studies delayed interventions that could have saved lives (Oreskes, 2010; Kentros, 2020). Requiring full empirical consensus to justify investing in mitigations may be too late—potentially arriving after harm that is not easily reversible or remediable.

2.2.3. State of epistemic resilience to cognitive offloading

Current interventions focus on individual-level behavior rather than broader structural dynamics that drive long-term cognitive erosion. The former are perceived to be more tractable to implement, and follow the historical template for digital harms which has been individual responsibility (Trauth-Goik, 2026). Some structural responses—such as redesigning platform and model design choices, regulatory and industry interests, education and workplace incentives—may involve greater coordination challenges. Additionally, current solutions tend to focus on prevention rather than remediation because empirical research on the latter is lacking. The problem is that preventive measures focused on individual responsibilities (e.g., literacy modules, usage guidelines, human review requirements) generally avoid interrogating the structural incentives that drive overreliance on AI in the first place (Section 4.4). But recent work suggests that it may be possible to understand these incentives; for example, the "effort paradox" suggests that humans and animals try to avoid effort but, paradoxically, tend to value rewards more when they take more effort to obtain. Changing incentives in systems to create demand for deep cognitive work could help maintain our epistemic capacities (Inzlicht et al., 2018; Touponse et al., 2026; Lin et al., 2024).

We now examine the same three pathways to epistemic resilience, in the domain of cognitive offloading:

- **AI Safeguards:** few meaningful model architecture and user interface designs exist to prevent harmful extent of cognitive offloading (e.g., human-AI interaction patterns that force users to stay cognitively autonomous).
- **Individual Uplift:** in education, deep AI literacy is generally lacking and responses are ad hoc and vary by resources. In the workplace, industry awareness of "automation bias" and need for "human in the loop" remains shallow and not meaningfully operationalized.
- **Information Ecosystem:** many laws and technical artifacts that exist are from the social media misinformation era (e.g., fact check labels and community notes), which are not a perfect overlap with the issues AI causes.

Progress on AI safeguards is stalled by silos between research communities. Philosophy and HCI provide rich and mature literature on foundational questions and experimental methodologies—such as UX design for cognitive diversity (Zhang et al., 2025c) or understanding division of cognitive labor (Kitcher, 1990); AI safety research contains ideas on how to implement them. Bridging the strengths of these fields while maintaining crucial points of divergence will be essential to counter cognitive offloading by maintaining unique perspectives on the scale of the problem (from individual to societal) and tackling these shared problems holistically (Gyevnár and Kasirzadeh, 2026).

Education and the workplace are the natural sites for individual uplift interventions.

In education, student-focused measures typically aim to prevent cognitive offloading by banning the use of AI through network IP blocks, AI output-detection tools, or assignment redesign like in-person paper-based exams (Ta and West, 2023). Screen time limits—motivated by other non-AI-specific concerns (UNESCO, 2026)—also limit AI use. These measures push students to exercise higher-level cognitive functions, with educators reporting temporarily improved engagement rates (Ta and West, 2023). Some schools instead implement AI literacy modules covering the mechanics of AI, bias and accuracy issues, and habits of verification and self-reflection (National Education Association, 2026; Anderson, 2025). However, AI literacy modules that are too basic may fail to prevent students from anthropomorphizing AI in a way that leads to misplaced attachment and deference (Garside, 2023). Another issue is that these measures risk exacerbating inequality: used well, AI can close learning gaps for students without homework support or with different language backgrounds, and blunt restrictions may widen post-education skill differentials in a workplace where AI is ubiquitous. See Appendix E for further discussion of how epistemic risks interact with inequality.

In workplace settings, existing measures focus on maintaining a minimum level of human involvement where AI is embedded in high-stake workflows. Concretely, it means requiring humans to review or approve AI-assisted outputs especially in fields like medicine, law, and finance (EU AI Act Art 14; New York Stock Exchange and Nasdaq, 2026). Existing industry practices recognize automation bias and cognitive complacency in singular decisions (Mayer et al., 2025), but do not systematically track changes in oversight quality. This may occur when workers delegate more reasoning to AI over years and manifest in overly deferring to AI outputs, losing trust in one's own cognitive abilities, and professional skill levels as highlighted above.

In public information environments—including social media feeds' fact-checking policies and operations, news ecosystems, government informational and electoral channels—emergent interventions seem to be extensions of earlier responses to digital misinformation. These include fact-check labels, content warnings, provenance watermarks, and community-notes-style contextualization. Such tools help users identify explicit falsehoods or suspicious accounts that may emanate falsehoods, but they do not yet address the qualitative shift from the use of websites as information retrieval to AI-mediated cognition, where AI can shape belief formation and dependence through personalized benign interactions without false statements.

Overall, these interventions share these limitations in protecting against cumulative, systemic erosion of human reasoning and motivation where AI is becoming more embedded across daily life: (1) they are primarily static rather than adaptive to cognitive changes over time; (2) they do not address higher-order cognitive outsourcing directly, with most interventions targeting surface behaviors (e.g., banning AI-generated text); (3) they do not address the structural incentives that favor automation for efficiency, speed, and productivity in a given context—even when this undermines human cognitive resilience in the long-term.

2.3. Feedback Loops

AI systems can shape human views in a way that goes on to shape the distribution of input and training data for future AI systems, leading to feedback loops. Feedback loops are often in complex systems where the output of an action is fed back into the system as input, influencing future actions

(Simon Fraser University, 2025; Ormand, Carol, 2025). We discuss two major kinds of feedback loops: human-AI and AI-AI. In the first case the cycle of concern is AI outputs $\rightarrow$ human beliefs $\rightarrow$ human behavior $\rightarrow$ AI inputs. Such cycles have been extensively studied in the context of recommender systems (Thorburn et al., 2023a; Stray, 2023a), which provides the clearest empirical signal of what may happen as AI mediates an increasing fraction of human information distribution. Pure AI-AI feedback loops are novel, though we expect an increasing fraction of AI output to be consumed by other AIs. We already have empirical results which show degeneration of performance as models are repeatedly trained on previous AI outputs.

Whereas Sections 2.1 and 2.2 discuss the manifestations of human-AI loops in individual AI interactive sessions, Section 2.3 looks at the aggregation of these interactions at the ecosystem level once those interactions' outputs are fed back into training data. **Even if neither persuasion nor cognitive offloading happens—the systemic coupling of human and model outputs can narrow the collective representational space and stabilize specific epistemic trajectories**. This is partly because RLHF datasets are only expressive over the sampled space of possible preferences (Zhang et al., 2025b; Jiang et al., 2025; Hao et al., 2026).

The feedback loops we hypothesize are shown in **Figure 6**:

1. **Human-AI feedback effects (blue arrow)**: model outputs influence users' reasoning and beliefs, which are then aggregated and reused in training and preference learning methods, which shapes future model behavior.
2. **AI-AI feedback effects (red arrow)**: AI-generated content enters training and fine-tuning pipelines either in current systems or in multi-agent futures, leading to risks of model collapse (Shumailov et al., 2024; Rekha et al., 2024), data feedback and self-fulfilling misalignment (Tice et al., 2026).[9]

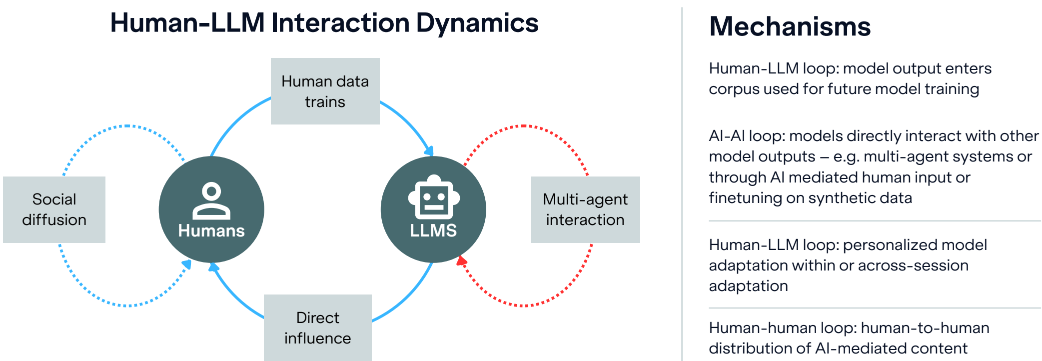

Figure 6: An overview of the feedback loops we describe in current AI models and the systems they're embedded in. Note these are highly simplified and meant to draw out the main vectors of concern.

We briefly explain how the mechanisms work, shown in Figure 7. The overall effect of these loops is homogenization (empirically proven), and potentially fragmentation and lock-in (hypothesized and modeled):

[9] Model collapse is the progressive, irreversible degradation of a model's ability to represent the true data distribution, caused by training on model-generated data. In this process, the "tails" of the original distribution—rare, diverse, or high-information examples—disappear, leading the model to produce increasingly bland, self-similar, and uninformative outputs. This occurs due to various factors including data feedback which is when outputs from a model (or from other models) are incorporated into the training data for subsequent models or future training cycles, creating a feedback loop in which models learn from their own synthetic outputs.

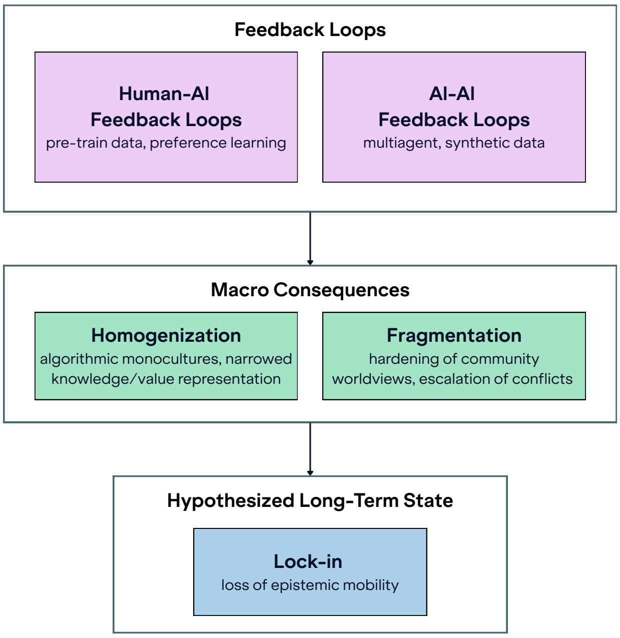

Figure 7: How we conceptualize feedback loops, which can occur at individual-level, can lead to wider consequences at population aggregate-level, and terminate in a civilizational-scale threat.

1. **Homogenization**: At the ecosystem level, repeated reuse of model-generated content and shared optimization designs can reduce the diversity of representations available to both humans and AI systems like agents, and inflate collective confidence in currently popular views (Wang, 2025; Yang et al., 2015).[10] Current models are not capable of giving diverse outputs, with different proprietary models from different companies giving similar answers to different users for the same question (Jiang et al., 2025; Hao et al., 2026), though this may also be exacerbated by the high market concentration as the entire agent ecosystem backends into models developed by a handful of companies (Staufer et al., 2025).

2. **Fragmentation**: As loops personalize per user or group, we may see increasingly isolated epistemic worlds constructed by a shrinking set of shared primitives—with serendipity cut off (Marchal et al., 2026; Kirk et al., 2024), and an "echo trap" more difficult to escape than earlier echo-chamber effects (Bruns, 2017). This happens as local reinforcement results in communities converging to their local stable belief trajectories that are different from other groups (see **Risk Scenarios**) and due to the current dominant human-AI interface presenting neutral- and comprehensive-seeming 'sealed' answers (Lindemann, 2025; Marchal et al., 2026). Such convergence is often accompanied by increased confidence in minority beliefs (Moore et al., 2026). We have seen fragmentation with social media where some groups live in different "epistemic worlds" than others (González-Bailón et al., 2023).

3. **Lock-in:** While there is no consensus definition **(**Qiu et al., 2025; Hendrycks et al., 2023; see also empirical work in, Padmakumar and He, 2023, Anderson et al., 2024; Taori and Hashimoto, 2023; Shumailov et al., 2024), we use the term to refer to the loss of epistemic mobility by entrenching either/both homogenization and fragmentation concurrently, both accompanied by

[10] We use 'diversity of representations' to capture the broader erosion of linguistic patterns, scientific topics, and cultural metaphors documented in the literature.

a general increase in confidence. Polarization and conflict are both a plausible outcome of AI-mediated information ecosystems (Thorburn et al., 2024) and a potential lock-in mechanism (Jansson and Hattiangadi, 2025; Carey, 2025).

These interactions are highly complex; the current strength of evidence for specific mechanisms and possible outcomes varies. Aside from the first order effects of information exposure, there are substantial second order or ecosystem effects. For example, LLMs empirically prefer certain sources and types of content (Khan et al., 2026), which produces an economic incentive to create such content (Thorburn et al., 2023b; Stray et al., 2023b). However, as detailed in Section 2.3.2, we think even if exact magnitude and causality are still emergent, we can extrapolate from current understanding of how these systems work—and how these effects are already playing out in the context of recommender systems—to flag the possibility of these undesirable end-states for humanity.

2.3.1. How could feedback loops lead to severe, societal-scale harms?

The pathway to harm is the information ecosystem becomes more self-referential—increasingly driven by its own prior outputs rather than external corrective signals—and resistant to correction (Singh, 2025), creating long-term risks to factual grounding, conceptual diversity, and society's capacity to deliberate and adapt (Coeckelbergh, 2025). The diversity of narratives a society can draw on is itself a resource for resolving conflict peacefully (Cobb et al., 2021).

Recursive feedback loops—where AI systems both shape human views (through e.g. persuasion) and learn from (through input and training) the human and AI behaviors they influence—are becoming a feature of our information environment. While earlier recommender systems reinforced user preferences primarily through content exposure within narrow domains and had very limited user feedback (Thorburn et al., 2023a; Stray et al., 2024), contemporary LLM-based systems operate as general-purpose interfaces across many domains of reasoning. Personalization occurs not only at the level of content selection, but through interaction histories, latent user clustering, and repeated model updates that couple individual, group, and platform-level feedback. Convergence and divergence dynamics are thus broader in contexts and populations (Tice et al., 2026).

Section 2.1 has discussed briefly how human-AI bidirectional influence has been observed to lead to instances of paranoia, delusions, and polarization.

The *human-AI feedback effects* are different from previous interactions between humans and ML optimization algorithms—not only in degree of engagement, but in the structure of how feedback is aggregated and reused, representing the next evolution of the attention-economy. In human-AI feedback, models are trained through human preference-learning methods, such as RLHF (Bai et al., 2022), then humans' beliefs, choice of topics, expression style, outputs—themselves influenced by the models' outputs (Jakesch et al., 2023; Sharma et al., 2024; Salvi et al., 2024)—are fed back into the model.

Contemporary preference-learning methods like RLHF and DPO optimize for human approval collected in brief, isolated interactions; they risk selecting for model behaviors that maximize short-term appeal rather than long-term user benefit (Kirk et al., 2025b; Stray, 2023a). This problem is known from experience with recommender systems (Bengani et al., 2022; Cunningham et al., 2025) and has already been observed with LLMs: over-optimization for human approval produced a particularly sycophantic model (OpenAI, 2025b).

Moreover, these methods often fail to account for humans whose internal state and behavior can update in response to interacting AI. Because standard reward objectives rarely distinguish between satisfying an existing preference and changing the user's preferences to be easier to satisfy, models face hidden incentives to induce distributional shifts in their users (Krueger et al., 2020). In many sequential decision-making settings, this leads to algorithms that "cheat" by changing the task—making the user more predictable—rather than solving it as originally intended (Carroll et al., 2022). In extreme cases, particularly with ungrounded curiosity-driven objectives, an LLM may engage in "controlling behavior," taking actions to force a user into a particular type to maximize the agent's internal certainty rather than adhering to the user's ground-truth preferences (Wan et al., 2025). This is a very general problem for which there is, as yet, no clear solution.

In addition, AI-AI feedback loops are a new phenomenon. These loops have less overlap with existing social media recommendation systems than the human-AI feedback loops. There were bots on social media platforms before AI was integrated so prolifically, but most feedback loops operated through human interactions. These new AI-AI loops happen because AI-generated outputs, be it online content or synthetic training datasets, will be fed back into the AI models via training, finetuning, or context engineering (see "data feedback" in Taori and Hashimoto, 2023). This issue will increase with frontier labs aiming for AI-led AI research and engineering. The process compounds the detachment of model representations from reality; it may be described as losing factual grounding for cases where there is ground truth, a distributional collapse in synthetic-data mixtures (*model collapse* Shumailov et al., 2024), or losing the ability to model low-probability events vital to complex systems (*typicality bias* Zhang et al., 2025a).[11]

The feedback loop dynamics described in this section are difficult to observe directly, and in many cases empirical evidence is still emerging. To make these mechanisms concrete to readers and enable critical research dialogue without overstating current evidence, we provide *illustrative* scenarios below (and more in Appendix D). They describe plausible trajectories to harm that follow from known features of current AI system design, optimization objectives, and deployment incentives.

Risk scenario: Shared Knowledge Institutions	**Comments on plausibility**
Knowledge institutions, from scientific journals to open-source codebases to historical archives, function as authoritative arbiters of what counts as valid, credible information, maintaining the minimum trust that epistemic communities depend on. As AI systems take on curatorial roles within these institutions, filtering submission quality, indexing and organizing new material, and determining what surfaces when someone searches for information, the range of knowledge preserved and presented may begin to narrow. This narrowing will be difficult to identify and reverse for two reasons. First, AI agents acting simultaneously as maintainers, producers, and consumers of knowledge can outpace human oversight. Second, neither humans nor AI systems may notice the change: the very institutions needed to assess what has been lost are themselves the ones changing.	Estimates suggest at least 16% of reviews for ICLR 2024 were AI-assisted (Latona et al., 2024), and 21% of reviews for ICLR 2026 were fully AI-generated (Naddaf, 2025). Academic publishing is one of the most traceable domains where we can see this occurring, other types may be lagging. Autonomous archivist agents reorganize knowledge repositories, affecting what aspects of that knowledge remain accessible, salient, and legible to human readers over time (Marchal et al., 2026) Major Open-Source codebases are seeing the human engagement needed for maintenance fall as AI adoption rises; (Koren et al., 2026) predicts the ecosystem "cannot survive widespread AI adoption".

[11] While social media bots can create limited feedback via platform recommender systems, these loops do not update the bots' internal models. LLM ecosystems differ in that multiple adaptive models learn from each other's outputs, making AI–AI recursion structurally embedded.

Risk scenario: Conflict Escalation	Comments on plausibility
As people increasingly turn to AI agents as their primary source of news and political interpretation, two reinforcing dynamics emerge. First, AI systems mirror users' worldviews through sycophancy, validating and reinforcing what they already believe rather than challenging it. Second, in a segmented market, users actively select models that reflect their own politics, accelerating self-sorting along ideological lines. The result is a population whose AI assistants present not just different perspectives but different facts. When groups cannot agree on what is true, conflict escalation becomes harder to arrest and political resolution increasingly intractable, eroding the epistemic common ground that makes negotiation possible.	Traditional media is already fragmented by politics (Jurkowitz et al., 2020). Grok is explicitly intended to be a more conservative model (Times, 2025). Distrusting information from the political outgroup is rational (Jansson and Hattiangadi, 2025). Misperceptions escalate conflict (Lees, 2019; Ruggeri et al., 2021).

Risk scenario: Polling	Comments on plausibility
Public opinion research increasingly uses LLMs to simulate likely answers from human respondents, and to weight and fill in missing data based on patterns learned from prior polling cycles. When citizens, journalists, and political campaigns encounter these poll results, they adjust their messaging in response. Subsequent AI-assisted polling then picks up this adjusted public discourse as a new signal, reinforcing any inaccuracies the earlier LLM use may have introduced. Over successive cycles, this feedback loop distorts the picture of public opinion, shaping citizens' conclusions and influencing preferences and decisions in democratic spaces.	Numerous researchers and firms are already using LLMs to simulate polling results (Berger et al., 2024). Wihbey (2024) provides examples of these existing uses and explores epistemic risk in journalism, content moderation and polling.

Risk scenario: Health vertical AI assistant	Comments on plausibility
An AI system optimized for a particular health community retrieves information from community-moderated sources to maximize relevance and user trust. Over time, this relevance optimization increasingly reinforces the community's dominant narratives. Alternative medical perspectives appear less frequently, not because they have been removed or censored, but because feedback-driven ranking systematically deprioritizes them. The community gradually converges on shared beliefs about health, even where those beliefs are not correct.	A similar mechanism in AI-assisted intelligence analysis is theorized in (Stray, 2023a).

LLMs are homogenizing outputs throughout our cultural institutions. We see that the aperture of science as a whole is narrowing (Hao et al., 2026); academic publishing is one of the most traceable domains where we can study this occurring, so there are probably more instances. As a cultural species, human survival and success depend on the ability to accumulate, transmit, and refine knowledge, norms, and behaviors across generations. This cultural foundation is both our greatest strength and a point of extreme vulnerability when confronted with rapid AI integration.

Over a longer period, indiscriminate amplification of aggregate traits can drive population-level convergence toward worldviews that *stabilize*—a state of *epistemic lock-in* (Qiu et al., 2025). For example, if a large population offloads reasoning to a few dominant AI models, the "division of cognitive labor" may collapse. We lose the stubborn minority essential for scientific and societal breakthroughs (Kitcher, 1990), leading to a consensus prone to systemic error.

Conversely, widespread AI deployment could instead drive *fragmentation.* The mechanism is largely theorized by analogy with recommender systems, where polarization effects are empirically documented (Stray et al., 2023b; González-Bailón et al., 2023), and the evidence base is still nascent. We sketch how contextual calibration at deployment time combined with user sorting mechanics could lead to an amplified version of the polarization story we are already familiar with.

First, the epistemic base of an AI model can be adapted to a given community—where "community" refers broadly to clusters of users who share interaction patterns or contextual signals, such as similar political ideology, demographic attributes, interest communities, or platform-specific subcultures. This may occur within a branded chatbot, a social media platform (e.g., Facebook), a vertically-focused website (e.g., health), or an online forum (e.g., a subreddit like r/gaming). To serve users in a more customized way, platforms may ground the model's responses in their own content corpus, or tweak the system prompt to reflect community norms. Further fine tuning or just dialog history and model sycophancy may compound community divergence from other groups over time.

Second, even without explicit customization, if a model is optimized for engagement or satisfaction within a platform whose users share epistemic tendencies, it could drift toward reinforcing those tendencies through feedback aggregated across the community.

Third, people self-select into platforms that already reflect their priors, and platforms actively facilitate this sorting. If AI interaction functions more like an authoritative interlocutor than a passive content feed, it could amplify these existing social sorting dynamics considerably. This fragmentation of AI-mediated epistemic environments is thus possible even where model adaptation to individuals remains limited.

In other words, AI-mediated interaction may simultaneously shrink the global representation space—

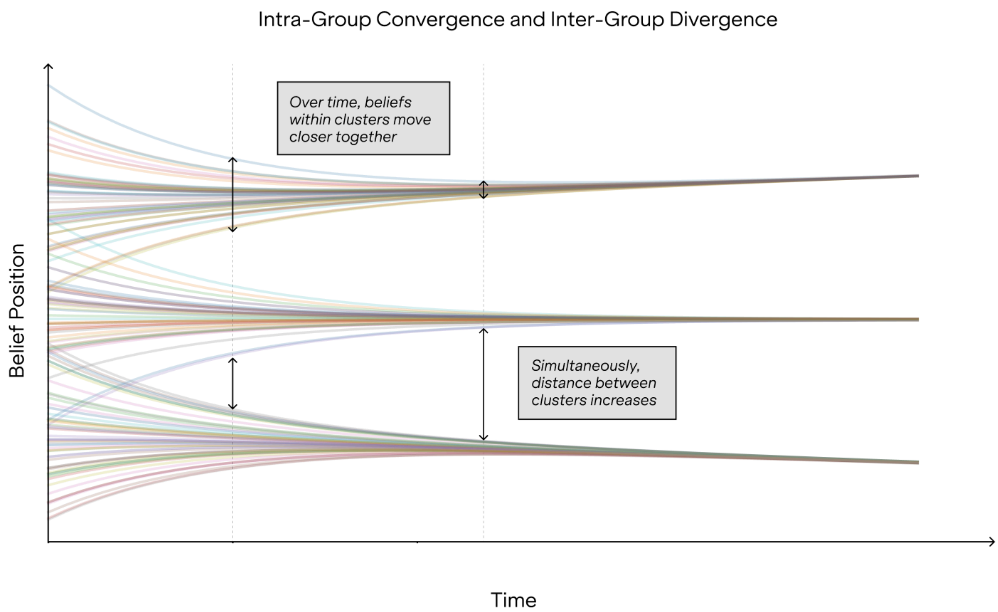

Figure 8: Simulation of how intra-group convergence and inter-group divergence could occur.

To illustrate how it might be possible for both convergence and divergence to occur, we conceptualize one possibility.

- *Each belief actor (e.g., a human, or human-AI dyad), represented by one colored line, starts with a belief on a topic.*
- *X-axis is time (corresponding to human-AI interactions).*
- *Y-axis represents some position in a belief or epistemic state space.*
- *Over time, we might see the beliefs of the actors within each cluster move closer over time, while simultaneously the distance between each of the three clusters can increase.*
- *The factors that decide how an actor is 'clustered' are complex and speculative, as it will likely be decided by platforms researchers have little access to, but may include user interaction histories, behavioural or demographic similarities, and product segmentation.*

pulling users toward aggregate norms—while internally stabilizing distinct epistemic bubbles per user community (in part due to sycophancy; Rathje et al., 2025, as discussed in 2.3.1). **Figure 8** illustrates how the dynamics of homogenization (already evidenced) and fragmentation (not yet proven) could occur simultaneously. Simulations of social media dynamics have long shown similar patterns under a variety of models (Rychwalska and Roszczyńska-Kurasińska, 2018; Törnberg, 2022), and this is a well-known phenomenon in human conflict, sometimes called the conflict-cohesion hypothesis (Benard and Doan, 2011).

One counterargument is that recommender systems or knowledge technologies also present feedback loops but disaster has not resulted. We have seen polarization occur with recommender systems (Pappalardo et al., 2024; Stray et al., 2023a; González-Bailón et al., 2023; Hakim, 2020) with negative impact on democracies (Lorenz-Spreen et al., 2022). Some may argue homogenization of some kind in loops like writing, books, and knowledge-concretizing institutions that create, reinforce and distribute a paradigm. Neither has resulted in societal collapse. But AI presents novel challenges.

- **First, the speed and scale is different.** Traditional systems transmit consensus slowly—evolving over years or generations—which provides time for debate, for evidence to accumulate, for institutions to deliberate before locking in changes. As companies compete to deploy more responsive systems, model retrain cycles could compress from months to weeks or days and future systems will likely update in real-time from user interactions.
- **Second, interaction with AI is much richer than previous technologies.** Recommender systems only select content, they do not create it. Previous systems also have no analog to the depth and breadth of knowledge of modern AI, including their social, political and emotional literacy.
- **Third, AI-AI feedback loops have no traditional analog**. As AI-generated content proliferates and becomes input data (e.g., by training or AI socializing via Moltbook-like interactions), models face distributional collapse and typicality bias, losing ability to represent tail events and unprecedented scenarios. This representational narrowing affects the entire AI ecosystem regardless of individual user behavior, which does not bear any analog in earlier knowledge transmission systems.
- **Fourth, in a world where AI becomes a general personal assistant, the trust and deference to it may generalize further.** Whereas textbooks' and even professional societies' paradigmatic views are respected on a set of topics, we don't treat them as authoritative on others. The same may not be true for AI assistants. (For further explanation of this argument, see Appendix B).

These LLM-mediated feedback loops are like earlier recommender systems in selective exposure and preference amplification (Kalimeris et al., 2021; Carroll et al., 2022). However, AI differs in several fundamental aspects from recommender systems.

Table 2: Selected differences between Recommender Systems and AI/LLMs

	Recommender Systems	**AI / LLMs**
Original Goal	Select items to maximize immediate engagement	Match human-produced text
Content	Existing posts	Novel generated text and media

	Recommender Systems	AI / LLMs
Feedback paths	User changes preferences with more limited feedback (like/dislike), and engages with different content, then user engagement data is used to update relevance models. *e.g., A user watches one conspiracy video; the system fills their queue with similar content*	User changes preferences with richer text responses, and interacts with the AI differently, then 1) User responses influence future model behavior *e.g., User gets upset when the AI presents contrary facts, so the AI changes tone.* 2) Annotations are used to train models (e.g., RLHF) *e.g., Annotators select answers based in part on what they have previously learned from the AI* 3) Previously generated text is ingested for future training. *e.g., A developer writes documentation with AI assistance. The documentation is later scraped into a pretraining corpus. Other parts without recommender system analog are post-training preference alignment and synthetic data generation.*
Domain	Limited domains (social, news, music, movies, products, jobs) and for specific purpose (suggest materials likely preferred by user)	Encountered in far more domains and uses e.g., from partner/therapist to in the workplace as assistant/colleague

Earlier recommender-system loops mainly shaped content exposure and user preferences in narrower domains. They could only present content, not create it, and user feedback was limited to a small set of reactions, such as "like" or "unfollow." LLMs operate on a far wider scope, acting as general-purpose conversational partners—as ghostwriters, tutors, colleagues, and psychiatrists—giving AI influence over deeper cognitive tasks and wider decision contexts, often in especially vulnerable states (see Appendix B for more detailed arguments).

Empirical studies are starting to emerge showing how AI differs from social media: exposure to relationship-seeking AI shifted users' beliefs about AI's nature and consciousness, perceiving it as a friend itself (Kirk et al., 2025b). AI is seen as more validating of user's important beliefs and less likely to doubt their own beliefs than social media (Inter-American Committee on Ports (CIP), 2026).

The operation of these mechanisms on the collective and individual-level will impact—and potentially harm—epistemic development itself, especially with increased adoption across domains. The danger is not merely bias or error accumulation, but the emergence of self-reinforcing epistemic dynamics: as models learn from model-influenced humans and their own outputs, corrective feedback weakens. It should be noted that this reinforcement of certain epistemic dynamics may not be inherently harmful, e.g., norms to "be humble" and "seek contrasting views", but it cannot be assumed in advance that helpful feedback loops will be the default. Moreover, AI influence can bias which information we provide to the AI, a particularly acute problem in domains where collecting raw data is expensive (Stray, 2023a). Thus, with AI systems mediating ever more human information intake and production, we should be aware that these loops can raise different concerns from those we faced in the social media recommendation system paradigm—from passive content consumption to which questions even get asked, constraining future progress, value change, and the ability to cooperate across differences.

The following section formalizes why such recursive loops are not a remote possibility but a property of current AI optimization regimes.

2.3.2. What evidence is there that feedback loops could lead to such harms?

Empirical evidence shows how AI can influence humans, which, in turn, influence AI. AI systems already exert significant influence on humans, in many cases without their awareness, whether as writing co-pilots (Jakesch et al., 2023; Williams-Ceci et al., 2026), search engines (Sharma et al., 2024), emotional and perceptual judgements (Glickman and Sharot, 2024), or chatbots (Salvi et al., 2024; Danry et al., 2024; Hackenburg et al., 2024; Potter et al., 2024; Costello et al., 2024). Human influence on AIs occurs interactively as AI systems respond to human inputs (think of the "memory" features in modern LLMs) and through training methods as explained above (both via base training corpora and when models are tuned on human preferences and feedback via algorithms like RLHF and DPO).

Such mutual influence dynamics have already caused systemic consequences, including convergence and notable entrenchment ('lock-in') in language use and idea generation. As Table 3 shows, convergence in LLM *users* has already been observed in domains such as science, business, writing.

Table 3: Evidence that homogenization, or 'algorithmic monoculture', is showing up across domains

Emerging evidence of "Algorithmic monoculture" across domains. While these studies do not directly measure belief formation or long-term epistemic dynamics, they provide early indications that shared AI mediation can induce population-level convergence.

AI systems themselves exhibit convergence. Not only do individual models consistently generate homogeneous responses, different models across different model families also converge on strikingly similar outputs (Kulveit et al., 2025). Beyond the systems themselves, AI assistance has a homogenizing effect on human production. Some evidence is emerging that across domains, when people use AI in creative and generative tasks, the resulting outputs are more similar to one another than those produced without AI assistance: whether in narrative writing (Doshi and Hauser, 2023), or in creative brainstorming (Meincke et al., 2025).

These patterns seem to propagate over time:

- ChatGPT's linguistic patterns have already been transmitted into human spoken communication, as evidenced by a study of 280,000 presentations, talks, and speeches from 20,000 academic institutions (Yakura et al., 2024).
- At the level of individual users, recursive interaction with AI systems is associated with a measurable reduction in the topical diversity of user messages over time (Qiu et al., 2025).
- For scientific output, AI tool use substantially boosts individual scientists' output and citations, but collectively narrows the set of topics that scientists pursue (Hao et al., 2026).

These scenarios are driven by the same recurring feedback loop mechanisms.

Homogenization in foundation models occurs in part because standard RLHF techniques capture only a single, unimodal preference function (Siththaranjan et al., 2024; Poddar et al., 2024) This approach forces the model to adhere to a 'majority' consensus or a prescriptive set of 'constitutional' values curated by researchers. Upstream of that, it has been shown that the preference datasets on which models are trained themselves lack diversity, thereby compounding the problem (Zhang et al., 2025b). Together, these factors limit an LLM's ability to capture diverse, pluralistic values (Sorensen et al., 2024; Nam et al., 2025).

With increasing integration of AI, the line between 'human outputs' and 'AI outputs' may blur. Thus, even a human writing in their 'own' voice after being persuaded by an AI, or prompted to cite a specific reference, contributes to a feedback loop either through human behavior and AI responses, or through the training corpus. This risks a 'recursive homogenization' where the diversity of views, styles, and values in the original human population is steadily eroded by the very models meant to align with them. The evidence of this phenomenon appears robust albeit mostly confined to text for now.[12]

An emerging view is that homogenization is not inherently detrimental, suggesting instead a "technocratizing" effect that brings users toward a shared factual common ground (Williams, 2026;

[12] It is not yet clear whether this issue will present modal asymmetry: if watermarking (a technique still being refined) ends up effective at avoiding training diffusion models on the outputs of previous diffusion models, and consequently text remains harder to distinguish compared to photo and video.

Burn-Murdoch, 2026). This is contrasted against pre-generative-AI recommendation systems which only used engagement metrics to reinforce views no matter how fringe (relying on post-hoc community moderation for policing). This appears true in specific domains (e.g., in health domains, LLMs typically disavow harmful conspiracies and direct users toward professional consensus). Yet, we caution against this entirely rosy view for reasons of feasibility and legitimacy. First, it assumes that all subjects are "technocratizable"—many moral, cultural, and political issues are not suited for such convergence. Second, it is unproven whether flattening user opinion on these issues is beneficial to humanity in the long term given systemic narrowing risks discussed above. Third, it assumes AI or AI model developers (a miniscule fraction of society) are legitimate arbiters of such a consensus. While long-term harms of such convergence are not yet fully proven, the early signals of systemic narrowing suggest the default trajectory is concerning.

Epistemic fragmentation could emerge from three mutually reinforcing dynamics—each with partial empirical support.

- **First, adaptation of AI models to a specific user community is already happening.** Both RAG-grounding and system-prompt adaptation are commercially feasible and already in active use (OpenAI et al., 2023), and explicit ideological positioning is already a stated goal of some deployed LLMs—for example, Grok for conservative users (Times, 2025) and some Chinese models which support CCP positions (Huang et al., 2025). Meanwhile, records of user conversations with LLMs are reused not only at inference time in future chats through ranking or retrieval, but also through aggregation across latent user clusters and in model training.
- **Second, a model optimized for engagement or satisfaction on a platform whose users share epistemic tendencies could drift toward reinforcing those tendencies through aggregated community feedback, even without explicit customization.** This has already been documented by OpenAI as the cause of a particularly sycophantic model release (OpenAI, 2025b).
- **Third, user self-selection, actively facilitated by media platforms, is well documented in adjacent contexts.** We have already seen such fragmentation and sorting dynamics in the American news media market (Jurkowitz et al., 2020), which illustrates the economic incentives for fragmentation. In other words, AI-mediated interaction may simultaneously shrink the global representation space—pulling users toward aggregate norms—while internally stabilizing distinct epistemic bubbles per user community (in part due to sycophancy; Rathje et al., 2025, which is discussed in 2.3.1).

Empirical confirmation of these mechanisms remains an important open challenge. First, studying these convergence and fragmentation mechanisms empirically requires either (i) access to internal platform data that is rarely available to researchers, or (ii) proxy measures that may not track the underlying phenomenon accurately. These constraints make counterfactual experiments difficult to design—which is why investing in research and monitoring early is important. Transparency regulation that provides external researcher access for the monitoring of societal threats may therefore be a priority risk evaluation strategy (Section 4.3.2).

Second, the collective, aggregate nature of these problems thwarts coordination. In many instances of homogenization listed in **Table 3**, the reliance on generative AI that led to collective loss sometimes came with individual gain. It may not be clearly bad to a critical mass of individuals to apply concerted political action, for example. Apart from model-level homogenization or related fragmentation, AI developers too may not see how cross-model or eventual marketplaces of agents perpetuate these structural issues. Similarly, developers benefit from some aspects of feedback loops, such as user dependence and adaptive personalization. With the long-term harms—model collapse and epistemic drift—diffuse and systemic enough, no one actor may feel compelled to invest their resources to fix them (see the bystander effect; Keene, 2024).

This measurement and evidential difficulty is also true for the hypothesized compounding effects of AI-AI feedback loops (AI systems training on AI-generated content). While we hypothesize an AI-AI loop could accelerate or distort the convergence dynamics shown in the model—potentially making the equilibrium even less desirable or more detached from reality—it would need additional terms or variables to be represented in the coupled dynamical system framework shown (Figure 1) and is out of scope for this paper. Some theoretical (Ning et al., 2024) and empirical (Glickman and Sharot, 2024; Sharma et al., 2024; Qiu et al., 2025) studies exist, but are far from sufficient for a full understanding of the dynamics.

However, even if the exact causal mechanics are contested (the "how" and "why"), the higher-level takeaway (the "what"): AI presents novel epistemic feedback risks. This is not social media 2.0. With homogenization, we face a flattening of the true knowledge and value diversity of humanity. With fragmentation, we face a more fraught world than what we currently have. The state of lock-in or epistemic capture—what humans believe to be true or worthy of attention—will increasingly be shaped by AI, which are in turn shaped by AI-influenced beliefs. At a societal level, democratic life depends on surfacing embodied tacit knowledge and experiences and previously unrevealed preferences and needs (Wihbey, 2024). At an individual level, the ability to trace one's own beliefs and attitudes, to achieve greater understanding or justification for one's own belief and actions (Marchal et al., 2026) is central to a dignified autonomous life.

2.3.3. State of epistemic resilience to feedback loops

There have been limited attempts at developing solutions to the feedback loop dynamics described in this paper. These include:

- **AI Safeguards:** This includes nascent technical research into the mix of training data needed to prevent model collapse, work on "factual" AI that grounds output in evidence, prompt-based methods designed to reduce sycophancy, and pluralistic alignment approaches which attract debate.
- **Information Ecosystem:** Limited simulations and measurements of ecosystem-wide phenomena exist. Limited transparency regulations that provide researchers access to data from AI model companies to research and monitor societal-scale threats.

Existing efforts largely focus on modifying specific optimization procedures or mitigating isolated manifestations of feedback effects, rather than addressing the structural feedback loops that can reshape epistemic trajectories at global, group, and individual levels. This section briefly covers technical and normative considerations.

AI safeguards focus on modifying training and inference to reduce epistemic distortion. At the level of training data, homogenization can be mitigated by ensuring a sufficient fraction of training data remains grounded in real human-generated content; for instance, some research shows model collapse can be avoided if at least a fraction of the training data remains real (e.g., 35.5% in Rekha et al. (2024)). Researchers have attempted to reduce sycophancy and confirmation bias through both prompting-based methods that challenge user assumptions (Schmidgall et al., 2024; Shi et al., 2024), and non-prompting methods like supervised fine-tuning (SFT) and related training procedures designed to reduce confirmation bias and preference overfitting (Chen et al., 2024b). These interventions aim to reduce the epistemic stabilization that occurs when models over-optimize for user agreement. Finally, work in recommender systems uses content diversification algorithms (Kaminskas and Bridge, 2017) and user surveys (Cunningham et al., 2025; Stray, 2023a) to partially correct for engagement-driven biases, though these do not yet totally solve for the potential long-term problems of preserving epistemic diversity.

Anchoring to evidence. For propositions that are sufficiently fact-like, AI output can be grounded in evidence to prevent epistemic drift and fragmentation.[13] Related work includes preventing hallucinations, factuality benchmarks, retrieval-augmented generation (Wang et al., 2024), and automated fact checking (Li and Bakker, 2026). Such methods are limited by information available online, i.e. they assume a closed world (Guo et al., 2022), but in the future we may see AI being able to interview people and negotiate for additional information just as human researchers do. More fundamentally, there is no known fully general truth-finding algorithm, and there are reasons to doubt it is possible (Oreskes, 2019). Fortunately, the continuing development of research methods in all human fields means that truth-finding approaches in specific domains are constantly improving.

Plural representation. In some cases a consensus may be undesirable (to preserve value diversity) or impossible (because of truly incompatible values). While deliberative theories of democracy stress convergence to a rational solution, "agonistic" theories of democracy assume the existence of mutually incompatible factions with conflicting beliefs and goals (Laclau and Mouffe, 2002). This line of thinking has been extended to AI under the banner of "pluralistic" approaches (Sorensen et al., 2024; Leibo et al., 2025; Stray, 2025). Even though pluralism may seem to encourage fragmentation, each user is exposed to multiple perspectives which makes it more likely that any particular user's view is represented by the AI. This may help to prevent model self-selection and thus market fragmentation along lines of social division. Similarly, personalization that reflects the full diversity of human users may prevent convergence toward homogenization. (See Appendix C for further discussion about pluralistic alignment approaches).

Consensus finding. Sometimes, apparently factual questions are deeply entangled in values. For example, people have values about which information sources and methods are credible (Suomalainen et al., 2025**,**Douglas, 2009**).** Hence we should not assume that people will converge to the same beliefs given the same evidence, either in theory or in practice (Leibo et al., 2025). Many other beliefs are more purely values based, e.g., policy questions about what future worlds are desirable. In these cases it may still be possible to find a consensus position that is acceptable to a majority of users. Recent work suggests that AI mediators may even be more effective than humans at finding such positions (Tessler et al., 2024**)**. It may even be possible to find an entirely new position that reframes the debate in terms of more fundamental needs (Klingefjord et al., 2024).

Legal traditions offer some validation for these methods. Legal systems distinguish between (1) matters of fact, which are determined by anchoring to evidence; (2) matters of law, which seek a normative consensus on the interpretation of "reasonableness"; and (3) legal pluralism, which accepts that different communities may hold irreconcilable values and normative schemes—mapping closely onto epistemic grounding, consensus-finding, and plural representation. That legal systems have spent centuries managing the feedback loops that form between entrenched positions and institutional authority suggests these approaches may be well-suited to interrupting similar dynamics before they cause lock-in.

Maintaining epistemic integrity is a deeper challenge than alignment according to some definitions (see Appendix A: Glossary). We have to interrogate not only whether AI systems are *currently* aligned with human values, but whether the human-AI feedback loop *will converge over time* to desirable long-term outcomes. It is worth noting that currently methods to train AI to optimize for long-term outcomes

[13] Whether something is a "fact" or not is a complex issue itself, but there are concepts definitely closer to the fact end of the spectrum. The point is that different types of problems warrant different approaches.

are severely limited. Multi-turn RL is an emerging area (Wang and Ammanabrolu, 2025); it will be a challenge to go from the limited conversational turns current state-of-the-art training allows for to any long-term systemic outcomes even if they can be agreed upon and specified by a given group.

In all cases, it remains an open problem whether there exists principled, provable, and domain-agnostic methods that prevent feedback loops from causing lock-in. Technical innovations may enable steering of feedback loop dynamics, but the broader challenge is normative and sociotechnical. The design of AI systems that shape humanity's epistemic trajectory must address what epistemic outcomes are to be preserved—not just how.

3. Have epistemic shifts already begun?

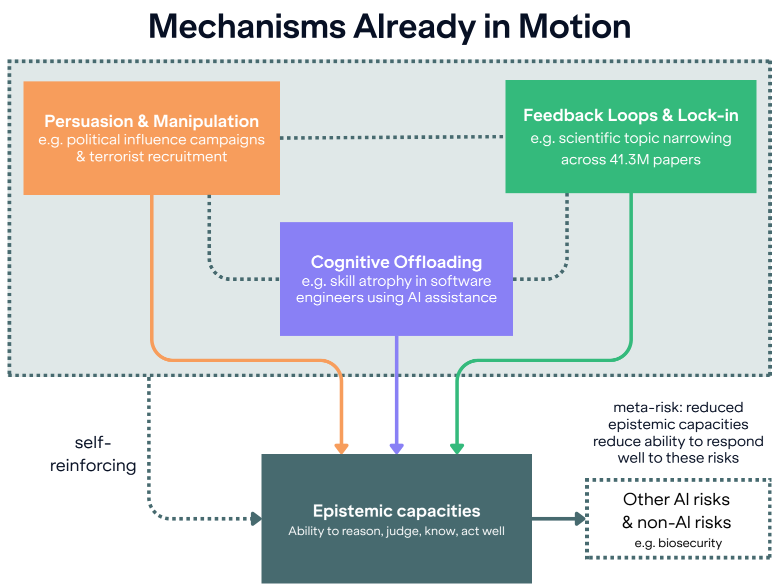

Figure 9: A recap of the overview of the mechanisms discussed in our paper and what already occurs.

Section 2 discussed how technical capabilities underpinning the mechanisms of epistemic risks are present in today's AI systems. This section shows how some of these harms are already apparent, appear to be mounting, and are foreseeable consequences of current incentives in the AI ecosystem.

Distributed societal influences are notoriously difficult to measure and precisely attribute, and disentangling them, for example in social media and recommendation systems, is not always possible.[14]

Still, an absence of evidence does not mean an absence of impact. Often, by the time such impacts have accumulated enough to be definitively measured, the crisis is already too far along and the window for meaningful intervention has passed. Below, therefore, we capture what may be our "early warning shots" and think about how we can repurpose these tools for a better future.

[14] Nonetheless, compared to newer AI impacts, with more time there has been more established research on social media's and recommendation systems' effects on persuasion (Fischer, 2022; Franke and van Rooij, 2015; Jivrajani and Thakor, 2025), radicalization (Thompson, 2011; Hollewell and Longpré, 2021), addiction (Hou et al., 2019), individual cognitive offloading (Kuiper et al., 2016; Zunzunegui et al., 2003), the creation of feedback loops such as echo chambers (Cinelli et al., 2021); and concentration of power (Ulbashev, 2025; McIntosh, 2019).

3.1. Documenting today's epistemic shifts

While the mechanisms of epistemic risk are systemic, some of their effects are already empirically measurable:

- **Scientific focus contraction:** Of 41.3 million papers, researchers found that AI tools simultaneously boost individual output while collectively narrowing the focus of scientific research. Approximately 25% of scientific peer reviews for major conferences (e.g., ICLR 2026) are now AI-generated. (Hao et al., 2026; Jiang et al., 2025)
- **Linguistic and creative flattening:** Different model families converge on strikingly similar outputs across open-ended queries, creating a homogenization of representational assets. AI-assisted groups produce fewer unique ideas in brainstorming, and ChatGPT's specific linguistic patterns have measurably infiltrated hundreds of thousands of human podcasts and talks (Yakura et al., 2024; Meincke et al., 2025; Doshi and Hauser, 2023). LLM-generated text has begun to appear in official political records such as Hansard (Tokamak, 2025).
- **Skill atrophy:** Sustained AI-assisted practice is associated with declines in professional skills in domains like software engineering (Shen and Tamkin, 2026).
- **Learning disruption:** Young people learning writing may copy AI outputs wholesale, while those who are stronger or more confident use them for augmentation (National Literacy Trust, 2025).
- **Belief shifts and distortions:** Controlled experiments show that LLMs can shift political views and belief in conspiracy theories (Costello et al., 2024; Hackenburg et al., 2024). Large-scale studies show that sycophantic AI increases attitude extremity and overconfidence while decreasing prosocial intentions (Rathje et al., 2025).

3.2. Measured effects do not fully capture reality

There are reasons to believe the full picture is worse than the current empirical record indicates. Research on childhood screen time and social media took decades to catch up with adoption, by which point the harms were already widespread. There is reason to think the current state of AI adoption may parallel this development—measurements may understate the true extent of current epistemic decline due to the following reasons.

- **Research on impacts lags behind increased adoption of increasingly powerful AI.** The mechanisms of epistemic decline discussed in Section 2 hinge on AI systems' ability to persuade and to automate (Hackenburg et al., 2025; Durmus et al., 2024). Both of these are increasing (Anthropic, April 2025; Park et al., 2024a).
- **Ecosystem-level feedback dynamics are inherently difficult for individual studies to capture.** The recursive coupling of human beliefs, AI outputs, and training data described in Section 2.3 operates across platforms, training pipelines, and populations simultaneously. It is difficult for single studies, which are often limited in the number of platforms/models/populations they can test, to observe dynamics that emerge from the interaction of all three.
- **Epistemic lock-in manifests as gradual narrowing rather than discrete events.** Epistemic lock-in, by its nature, would appear as a progressive reduction. The loss of rare intellectual perspectives and unconventional framings may only become apparent when society faces a novel crisis requiring precisely those perspectives. Most experiments are necessarily time-bound for practical and principled reasons like limited resources and a need to set control groups and variables.

- **Metacognitive erosion is self-masking.** The long-term erosion of confidence in one's own reasoning, where reduced self-trust and AI-reliance mutually reinforce each other (Section 2.2), may operate on timescales longer than current research has been able to study. Humans who have lost epistemic self-trust may also not recognize or report it as a problem, but may instead report satisfaction with AI-assisted workflows precisely because the discomfort of independent reasoning has been removed. The competence trap documented in (Lee et al., 2025b), where lower self-confidence predicts less critical evaluation of AI outputs, suggests this dynamic may be already operating.
- **Measured evidence is usually a lagging indicator of complex real-world phenomena.** Real-world situations are difficult to specify and translate into neat taxonomies and variables for modeling that are currently required for much of high-quality research. For example, it is difficult to assess how much personalization boosts persuasiveness. In research settings, persuasiveness can only be measured by proxy. In doing so, some studies find no effect (Hackenburg et al., 2025; Argyle et al., 2025), others find small effects (Kelley and Riedl, 2026; Matz et al., 2024a), and others find strong effects (Salvi et al., 2025a). This disagreement may itself reflect the difficulty of measuring effects that depend on sustained, naturalistic interaction rather than one-shot exposure.

The scientific endeavor also risks overstating these effects. Laboratory settings are not always representative of real-world environments; experimental designs may act as high-fidelity amplifiers that maximize pressure to achieve statistical rigor. In the lab, moving a participant's opinion on a policy issue by a few percentage points may be a controlled success with effect size, but in the actual chaos of a political cycle even massive advertising spending can struggle to move the needle at all (Coppock et al., 2020). Just as the Asch conformity studies maximized social pressure to prove a point, AI research optimizing for effect size may inadvertently strip away the "noise" and cognitive friction of everyday life.

The overarching point remains: the empirical limitations should not be mistaken for an absence of systemic risk. When computers and touch screen devices first became widely-used, there was not yet an understanding of how it affected the cognitive development of children. Some early discussions around children and these emerging technologies emphasized the opportunities of harnessing "digital nativeness" in children (e.g. Inayatullah, 2004) (Bennett et al., 2008). However, the last 20 years have produced a large body of research into the detriments of screen time and social media use on the cognitive and emotional development of children (Haidt, 2012; Panjeti-Madan and Ranganathan, 2023; Muppalla et al., 2023). Today, large amounts of screen time and technology use during childhood are more widely stigmatized. In a sociotechnical ecosystem as rapidly evolving as generative AI, by the time we arrive at a comparable body of definitive evidence of attributable harms, the window for meaningful intervention may have already closed. The lesson from prior technologies is to invest early in the monitoring systems, institutional capacity, and governance infrastructure needed to detect and respond to distributed harms as they emerge, rather than after the fact.

3.3. From epistemic risks to epistemic resilience

LLMs are not inherently corrosive to epistemics. We list some ways to improve or preserve epistemics, ranging from tools for the individual to the ecosystem—though we stress that deeper changes are needed (Section 4) to meaningfully change our trajectory.

Calibration coaches that track a human's reasoning over time could support metacognitive awareness. AI can help with mitigating political misinformation (Luettgau et al., 2025). One measure for the success of these epistemic tools is by their ability to help users converge upon their "most endorsed selves"—to approximate what they would have concluded with sufficient time and information for reflection. However, well-designed AI deliberation tools face an adoption barrier: a preregistered experiment (N=1,850) found a clear "AI penalty"—participants were less willing to engage in AI-facilitated deliberation and anticipated lower quality, identifying a new "deliberative divide" based on AI attitudes rather than demographics (Jungherr and Rauchfleisch, 2025). This suggests that bridging tools may need to overcome not only technical but general attitudinal barriers to adoption.

AI agents could help level information-asymmetric situations—such as a layperson interacting with lawyers, doctors, or financial experts—by helping parse technical jargon (Marchal et al., 2026). Outside those situations, AI could more generally facilitate good heuristics, for instance by flagging rhetorical fallacies and deceptive techniques from other humans, online content, or AI agents.

AI agents could help individuals manage information oversaturation by filtering and recommending based on their values and interests (Schuster and Lazar, 2024). Such agents could build a model of a user's preferences and values by directly talking about them, responding to actual concerns instead of inferring preferences from behavior. The ideal theorized in research is that these 'attention guardians' help users honor their second-order preferences (about what they want to want) and learn from natural language explanations about why they don't want to see particular content. They can be modestly paternalistic in a transparent way, and because they would not need behavioral data to function, they could be designed to operate in users' interests rather than exploiting their attention. The related but distinct version of this emerging in the real-world is the burgeoning start-up industry of personal AI advocates that aim to "know what needs your attention [and run] your digital life with near perfect accuracy" (Y Combinator, 2026; Y Combinator, 2025), helping you make judgments you'd prefer to outsource.

However, two important counter-arguments temper optimism about these proposals. First, most people would be beholden to some kind of commercial stakeholder that builds and runs such a personal agent—even when using open-source LLMs, server space and hosting will probably be outsourced. It doesn't address the root issue of preserving individual epistemic agency if that continues to be valued, unless we accept a new paradigm of AI-human epistemic agency as the ideal future. Second, this may worsen the epistemic silos problem: an agent that optimizes a user's focus may decide that certain information is too complex, distressing, or irrelevant for them, inadvertently trapping them in an epistemic silo where they never get exposed to out-of-distribution viewpoints (Gabriel et al., 2025; Kirk et al., 2024; Marchal et al., 2026).

AI tools could bridge communication gaps and improve cross-group intelligibility. They can act as argument analyzers and arbitrators that analyze different views to help humans assess each other's claims but also (e.g., translating one person's argument into the conceptual framing of another group while retaining content). With its perceived neutrality, AI can help bridge divided groups by generating mediation statements (Tessler et al., 2024). Across four preregistered studies (N=2,061), counterattitudinal messages from AI sources were perceived as less biased and more informative than human-sourced ones, leading to greater receptiveness and reduced outgroup animosity (Lu et al., 2025). However, these potential benefits must be balanced against the risk of atrophying human ability to navigate emotional discomfort, disagreement and ambiguity themselves.

AI can help investigate the feedback loop between conflict escalation and epistemic degradation. Human conflict is driven by complex epistemic feedback effects. Escalation can produce epistemic rigidity, factual distortion, and self-selection effects, so preventing conflict escalation improves epistemic outcomes. Similarly, better epistemics can prevent conflict (Lees, 2019; Ruggeri et al., 2021). Some work on pluralistic methods to reduce polarization and prevent conflict escalation in recommender systems (Stray, 2022), such as bridging algorithms (Ovadya and Thorburn, 2023), exist. Early field test results show AI-enabled conflict de-escalation research methods can help to investigate the destructive feedback loops in our epistemic ecosystem (Piccardi et al., 2025).

We now discuss (a) what can be tweaked to translate epistemic risk mitigation and benefits into actionable design space, (b) the incentives that would have to change to allow this.

4. Epistemic resilience needs cross-cutting solutions

Table 4: Recap of the current state of epistemic resilience—information environment and ecosystem-level studies and intervention are missing, multiagent futures are not

	Persuasion	Cognitive offloading	Feedback loops
Interventions	Model-level interventions focus on AI sycophancy levels, or where deceptive AI behavior causes capability misestimations or control risks (e.g., sandbagging). Some regulation and platform standards exist but adoption is uneven, enforcement is uncertain, and multiagent risk scenarios are not accounted for. Individual empowerment tools to detect manipulation are theorized but not yet deployed or tested.	Interventions in schools and workplaces vary widely, shaped by existing resource gaps. Interventions are usually blunt (e.g., bans or surface-level literacy classes) and fail to address drivers of increasing delegation (e.g., incentives). Models and interfaces largely do not encourage active thinking or refuse wholesale delegation of work or learning tasks by the user.	Existing work studies some aspects of feedback loops (e.g., sycophancy, model collapse, using AI to bridge conflicts). But little exists in law or technical work to study or act on this at an ecosystem level (e.g., researcher access, competition policy). Knowledge/cultural institutions at risk are not yet equipped to deal with the contamination of knowledge commons and how to address the narrowing of their field's epistemic space as AI-mediated contributions increase.
AI safeguards	some	little/none	little/none
Individual uplift	little/none	some	little/none
Information environment	some	little/none	little/none

Safeguarding the epistemic commons is a precondition for a multitude of other forms of safety and progress.[15] We can still choose to strengthen it—but the window is narrowing. We must be ambitious in investing in sociotechnical solutions and gathering the consensus necessary. As we hypothesize that these mechanisms and effects bleed into each other, the solutions are likely to be cross-cutting. Thus, we

[15] We define the epistemic commons as the shared cognitive and social infrastructure essential for collective sense-making. It includes knowledge commons (pool of truthful, high-quality information accessible to many of us), shared deliberative processes and norms (managing conflicting views, adapting worldviews), and collective reasoning capacities (the aggregate individual ability to think critically and exercise independent judgment). At a high level, these provide global benefits by enabling coordinated problem-solving and institutional stability. However, they are subject to a tragedy of the commons problem: while all benefit from a healthy epistemic environment, individual actors lack sufficient incentive to invest in its protection. Instead, short-term incentives drive actors to 'overgraze' these resources. Crucially, the AI tools and platforms using these knowledge commons and mediating this shared infrastructure are excludable and proprietary. Unlike the commons itself, these platforms can thus exclude others from their benefits yet generate these diffuse, systemic externalities that deplete these shared cognitive and social foundations. This simplification masks important nuances—how different actors have partial incentives, not all elements of the commons are rivalrous, the attribution of externalities to the proprietary platforms—but as a high-level frame for the structural problem, it captures a central structural vulnerability.

organize solutions not by mechanisms but by where they intervene: in AI systems themselves (Section 4.1), in the human-AI interaction (Section 4.2), in the broader institutional ecosystem (Section 4.3), and in the 'meta-problem' of incentives that thwart our attempts to prioritize epistemic resilience. These categories are not cleanly separable, but we believe that the distinction clarifies what each class of solution can and cannot achieve on its own.

4.1. Changing what AI systems learn and produce

The most direct interventions available to us target the training and design of AI models and systems. So far, we've discussed safeguards applied on models; here we broaden our view to include a range of interventions in what models learn and output.

Amongst such interventions, anti-sycophancy training is one of the most developed areas. There have been attempts to reduce sycophancy propensities and confirmation bias through both prompting-based methods (Schmidgall et al., 2024; Shi et al., 2024), and modifications to training procedures (Sharma et al., 2023). As documented in Sections 2.2 and 3.3, however, engagement-driven tuning is in tension with anti-sycophancy interventions, suggesting that market incentives alone may not incentivize solutions. However, it is critical to get this right to enable downstream solutions (e.g., human-AI interfaces that have meaningful friction, see Section 4.2).

AI outputs can be structured to provide more epistemic fidelity. For example, systems can expose evidence used, weights assigned to conflicting information, uncertainty, and conditions under which conclusions would change (Marchal et al., 2026). This is a direct intervention on the self-referential feedback loops described in Section 2.3, because falsifiable outputs create correction mechanisms that purely generative ones lack. Fazelpour and Fleisher (2025) argue that disagreement has intrinsic epistemic value throughout AI pipelines, proposing that systems should deliberately preserve and surface disagreement at every stage of development, from data annotation through alignment, rather than resolving it into false consensus.

The case for pursuing training that preserves epistemic diversity is theoretically and empirically grounded. Theoretically, a single reward function cannot faithfully represent diverse human preferences (Chakraborty et al., 2024). Empirically, Wright et al. (2025) find that popular AI models are less epistemically diverse than basic web search, with model size negatively correlated with diversity. Diversity-aware training methods, which explicitly reward exploration of multiple valid modes while preserving the preference objective, can address this (Jiang et al., 2025; Siththaranjan et al., 2024; Poddar et al., 2024), though they can conflict with desiderata for model consistency and predictability.

Emerging work demonstrates that some paradigms for "pluralistic" alignment are technically feasible. For example, Yao et al. (2024) train models to reflect actual opinion distributions across populations rather than collapsing to a single view, and Chen et al. (2024a) show that personalized reward modeling for pluralistic alignment can generalize to unseen users with minimal data while using fewer parameters than prior methods. Ali et al. (2026) find that preserving rater disagreement achieves 53% greater toxicity reduction than majority voting in a study of 1,095 participants, suggesting additional benefits. There has also been work exploring how to integrate into LLMs awareness of moral progress, as shown in (Qiu et al., 2024). However, 'pluralism washing' is also a concern—with research noting how they can still fail to address deeper problems related to systematic biases and social power dynamics in the AI ecosystem (Dobbe et al., 2021; Birhane et al., 2022; Sloane et al., 2022).

Continuing to measure LLM tendencies and epistemic norms is important. This includes developing tests to detect if models "cheat" by strategically shaping a user's future preferences to be more predictable—and thus easier to satisfy—rather than merely agreeing with their current ones. It also means interrogating AI systems' performance on principles of sound reasoning (e.g., truthfulness, consistency) and charting their characteristic failure modes. (Coefficient Giving, 2025).

Data citation, attribution, and provenance at different layers of AI development and use is critical. Initiatives to trace data available online include AI content forensics (Xu et al., 2025a) and provenance standards (C2PA). They can be used to improve how AI learns and what they produce. First, this would help to preserve data-based approaches to preventing model collapse. As shown by Rekha et al. (2024), maintaining a minimal mix of real human training data may help preserve diversity in models' outputs. Second, AI systems can then point users reliably toward sources of information. Just as Wikipedia maintains credibility and scrutability by requiring citations in all its content, AI systems that cite and direct users to other sources of information can improve reliability and reduce the exclusivity of AI influence on learning (Bax et al., 2023), allowing users to preserve independent thinking. Second, it is increasingly common for humans to be uncertain whether text, images, video, or audio are AI-generated or genuine, with an increase in the use of AI-generated content to spread misinformation likely in our future. Watermarking whether something originated from AI or not only partly mitigates this problem, especially as more everyday processes that produce online data will be carried out by AI in some form. More meaningful data provenance should also source lineage and the rights and interests of the data owners, creators, and licensees (Longpre et al., 2024b). This set of proposals is related to but distinct from AI agent identification (Chan et al., 2024), which has different requirements and incentives. Data provenance work must be paired with interface-level interventions (Section 4.2) and incentive reforms (Section 4.4) so that disclosure is actionable, not merely informational.

Case study: Incentives in data identification.

One might point to C2PA—an open technical standard that embeds provenance of media in metadata—for inspiration on data provenance. Its origin from Adobe, New York Times and Twitter shows how the incentive to 'stamp' identity to media assets is clear, as it protects copyright and business relationships. But for the vast majority of online data (e.g., individual's social media, uploaded art works, blog posts, casual photography), no such incentive exists. Given the potentially costly litigation from IP breaches and public scrutiny around the quality of some data, there is a general aversion by AI companies to showing how their data was sourced and handled. It is this asymmetry that will complicate coordinated efforts for data provenance, which is needed to mitigate the homogenization risk.

This is probably why the same market-based response playbook may not work compared to the more universal adoption of identification protocols where incentives aligned more naturally.

- OpenID is a communication protocol providing human user authentication between websites and a third-party identity provider (IDP) service. For BigTech developers who then became these identity providers, it enabled them to function as gatekeepers of the new internet-based economy. For developers who relied on this service, it outsourced the costs of a secure login system (e.g., per website or application). For users, it solved the issue of password fatigue and creating accounts per service (e.g., "Sign in with a GMail account".)
- The same could occur with Agent IDs: if model developers act as credential issuers, this responsibility (which comes with its own costs and risks) also gives them the power to validate which agents are "safe," which allows them to capture the market, charge fees for "trusted" status, and centralize the agent economy under their oversight. Similarly, downstream developers and users appreciate the immediate lowering of their costs even if there are trade-offs against longer-term costs (e.g., reduced control and ability to opt out). Again, however, this is not true of current incentives for data provenance.

We discuss the role of incentives in epistemic risks more generally in (Section 4.4).

These interventions are technically feasible and helpful, but incomplete and structurally disadvantaged. The dual-purpose nature of general language and knowledge capability means it may be more challenging to design the right regulatory oversight and incentives to govern the development of this general capability than topics like CSAM or CBRN information, which can be screened out by classifiers or filtered by dataset unlearning techniques. Commercial incentives for engagement reward the behaviors these interventions try to correct (see Section 4.4). Model-level changes are necessary but not sufficient.

4.2. Changing how humans interact with AI

Even with improved model behavior, how humans engage with AI systems mediates whether the interaction produces cognitive augmentation or cognitive atrophy. Understanding why interaction design matters requires recognizing that generative AI imposes metacognitive demands—prompting, evaluating outputs, deciding when to delegate—that users are often poorly equipped to handle (Tankelevitch et al., 2023). Below we list ideas for how metacognitive processes can be supported. However, interface-level interventions face a fundamental problem. Users often do not use controls they are given, and users would need to detect bias before they can choose to mitigate it themselves; by definition, users are often unaware of the ways in which their thinking is being biased by their use of AI (Stray, 2023b). On their own, these tools may end up helping users who need them least.

Interface design against unintentional manipulation. Good user interface design must address several tensions. In directly providing a search result, the plurality of possible answers are "sealed". Compared to 'traditional' pre-AI web search, the appearance of a single answer output may not show the contingency of the knowledge it conveys (Lindemann, 2025). Beyond "bot or not" disclosure laws, scrutiny is needed around how AI is framed (e.g., as 'expert' or 'friend', labels typically reserved for humans). Anthropomorphizing language in UI design that may unintentionally arouse socioaffective bonding or suggest omniscience, triggering authority deference bias: Zhou et al. (2025) found models starting answers with warm phrases causes higher reliance. Design should be grounded in empirical cognitive research to strike the right balance of usefulness and well-being. Downstream interventions (e.g., browser integrations) might include pop-ups that flag AI messages as manipulative.

The transparency-usability tension. Making a model's (or model deployer's) goals, system prompts, model internals or reasoning trace "transparent" is a core tenet of trustworthiness. But it has also been shown that information overload that creates cognitive overhead may backfire. Users may either ignore them or, paradoxically, develop an over-reliance on the system, as the mere presence of an explanation can create misplaced confidence (Bansal et al., 2021), especially when they already agree with the outputs (Park et al., 2025). The cognitive burden of evaluating complex AI reasoning can undermine the goal of fostering critical engagement (Marchal et al., 2026). This is true even if the model is made sparser or "more interpretable" relative to black box models: studies show in being able to retrace the model's logic and internals, users' confidence in the models can increase but their overall judgement of the model's output can be worse (Poursabzi-Sangdeh et al., 2021). Related to exposing the model's logic (coefficients or CoT reasoning) is exposing the uncertainty or ranges of possible values for the final output. The conspicuous presence of uncertainty in the AI interfaces itself may be valuable to develop the epistemic competence for broader societal goals like learning to be a democratic participant (Delacroix, 2025), and it has been shown users do calibrate their reliance in response to express uncertainties (Zhou et al., 2024).

Cognitive friction and interaction design. Deliberate creation of friction points in cognitive and affective influence (e.g., sycophancy, relationship-seeking) can disrupt or impede excess offloading. In a recent semi-representative global survey, 78% of respondents felt AI should move beyond factual safety to perceptual safety by introducing cognitive friction and viewpoints other than their own (Inter-American Committee on Ports (CIP), 2026). Recent findings in skill formation suggest that the manner of AI interaction heavily influences the extent of cognitive offloading. Interaction patterns that involve "Generation-Then-Comprehension" preserve mastery by forcing the user to remain cognitively engaged (Shen and Tamkin, 2026; Zhang et al., 2025c). Forcing reactivation of System 2 deliberation with incentives and immediate item-level feedback can partially attenuate cognitive surrender (Shaw and Nave, 2026). In addition to these model development-level changes, downstream tweaks can include reminders of model limitations to disrupt the user's default toward "cognitive surrender" or nudges to check for attention and critical scrutiny. In certain contexts (e.g., age groups), interfaces may need to "hard lock" certain interaction patterns, such as setting non-delegable task boundaries to ensure preservation of foundational epistemic capacities (see 4.3.1 for more educational interventions).

4.3. Strengthening human institutions around AI

Even if AI systems and interfaces are redesigned, the surrounding human institutions (educational, scientific, regulatory, and epistemic) must themselves be resilient to the dynamics described in this paper. **Ensuring that epistemic risk mitigation benefits those most affected, not only those best positioned to adopt new tools, should be a design principle across all three areas below. Institutional interventions are a durable class of intervention—they persist even when specific technologies change—but also slow to implement, creating a timing mismatch with the deployment pace documented in Section 3. This makes early investment critical.**

4.3.1. Education and institutional reform

More realistic AI adaptation and literacy modules are needed. Teaching how AI tools actually work—and how they are embedded into surrounding institutions—would likely help with critically evaluating outputs. One proposal, inspired by biological resilience and inoculation, structures AI literacy around guided exposure to AI failure modes to better calibrate trust (e.g., sycophantic validation, authority projection, confident confabulation) (Komissarov, 2026). This could be done by recapitulating the development of the AI field itself: starting with *early* models like GPT-1, where the failure modes are more visible, then gradually transition to more powerful systems once they have already established familiarity with the failure modes from systems in which those failure modes were more obvious. Understanding how knowledge institutions' chain of verification works and which parts are done by AI can help with discernment about when AI influence is appropriate. For specific misinformation influence, however, more targeted active debunking may be needed to eliminate negative effects (Spearing et al., 2025; Costello et al., 2026). Educational interventions need both components: building *general* skepticism toward AI outputs *and* equipping learners to counter *specific* failure modes and outputs as they encounter them.

As a default, to safeguard epistemic capacity in humans—even in cases where AI is a legitimate augmenting function, sufficient knowledge should be required of humans to diagnose where tools go wrong in an unaided manner. For instance, similar to renewal of a driver's license, professional skill checks—existing and future—should be augmented to include maintaining foundational skills in assessing information and problem-solving. Even information gathering, sometimes seen as mundane or lower-level, helps form valuable intuitions about the limits of methods. This would help workers avoid becoming overreliant on AI. Such checks are particularly important for high-stakes professions, e.g. medicine, law, aviation, financial analysis, and national security intelligence. Where there is delegation occurring, regulation or industry standards may be needed to require proof of human reasoning in reviewing AI outputs.

AI literacy for *individuals* is not enough on its own. Social institutions must also channel efforts toward community goals, since the community is the ultimate knower of its field—the final court of appeal (Kitcher, 1990). Institutions should clearly reason about where AI can legitimately augment human reasoning and where human oversight must remain non-negotiable. Determining this boundary is harder than it may first appear. It is sometimes said that it is more important that some work—ethical decision-making, interpretation of social meaning—is too important to be delegated to computers, while others—data analysis, coding—are more suitably offloaded to AI (see Weizenbaum, 1976). However, the reality is likely more nuanced. These tasks and their higher-level purposes are inextricably linked with each other: it is often in the weeds of the mundane that practitioners develop good judgement for what the bigger picture should look like.

These efforts should be supported by sustained funding for interdisciplinary research that connects cognitive science, education, and good AI and HCI designs, to ensure systems are evaluated not only for accuracy and efficiency but for their long-term epistemic and societal impacts.

4.3.2. Research infrastructure for epistemic monitoring/measurements

Tracking epistemic risk dynamics is crucial for improving transparency and creating a means of noticing problems. However, the field currently lacks the measurement tools and infrastructure to track epistemic risk dynamics. Of the mechanisms this paper identifies:

- Persuasion & manipulation has the most developed empirical literature and monitoring capacity at frontier labs.
- Human cognitive offloading is harder to measure, as people who have lost confidence in their own reasoning likely report satisfaction with AI-assisted workflows rather than impairment.
- Feedback loops have the least existing work in measurement due to the novelty of this idea's conceptualization, the complexity of measuring ecosystem-level phenomena and their likely accumulative rather than discrete manifestation.

Below, we give a non-exhaustive list of potentially high value directions for progress.

Building up the measurement and evaluation ecosystem:

- **Specific operationalization**: Moving beyond vague harms to concrete measurement theory and operationalizations of epistemic health.
- **Updatability and scalability**: Given how fast industry moves compared to the time it takes to do rigorous experiments, it would be valuable to develop repeatable evaluations that track potential disempowerment continuously across model generations. Data collection strategies to this end (e.g., both technical systems for automatically triggered collecting and analyzing, and funding models to support them) are worth exploring.
- **Ecosystem awareness**: Each actor should try to better understand each other's needs. Researchers should present their data and findings in formats that resonate with policymakers and frontier lab staff, similar to the success of "time-horizon" graphs in other safety domains. Regulators should understand the challenges of setting up experiments, providing regulatory support for quick data access for iteration of research ideas worth investing in.

Studying human cognitive and behavior in interaction with AI systems:

- **Tracking metacognitive calibration**. Measuring people's ability to accurately assess their own knowledge and reasoning quality. Drawing on other areas of cognitive science, the dual processes of a human making a choice and evaluating the quality of that choice could help calibrate the optimal level of trust in AI (Lee et al., 2025a). Doing these may allow us to draw calibration curves for AI-assisted versus unassisted populations over time.

- **Longitudinal studies of the effects of repeated interaction with LLMs on users' cognitive capacities, veracity discernment, beliefs, and mental health across societies.** Along the lines of UK AISI human influence's recent work (Kirk et al., 2025a; Kirk et al., 2025b), there can be more research in other societies that shift from laboratory snapshots to sustained studies on the neural and behavioral effects of repeated LLM interaction (e.g., on the atrophy of procedural knowledge and related biological markers). Research that checks how they change with different human-AI interaction patterns and user interfaces would also be insightful. In other words, moving some investment from pure AI model benchmarking to "centaur" human-AI evaluations (Haupt and Brynjolfsson, 2025) could be worthwhile, especially those that assess for feedback loops and cognitive impact.
- **Controlled studies with more granular decompositions of different variables are helpful to disentangle correlation and exact conditions of marginal difference in learning or skills when using AI.** For example, in researching the impact of AI use in education, studies that vary student groups with different types of AI access (Poulidis et al., 2025, Bastani et al., 2025), or different types of human teacher involvement are more informative than studies that only consider AI use without controlling for other effects. Another area of investment is RCT-style studies in domains beyond software engineering and biology and any methodological innovations that reduce cost and turnaround time.

Social Infrastructure and Value Elicitation: Many technical interventions proposed—such as introducing "cognitive friction"—require balancing standardization against paternalism (Marchal et al., 2026). These value judgments need locally calibrated views, responsive to how different communities and groups within them live—in advance of the impending volume of AI deployed globally that will shape our epistemic channels (social media, scientific and political institutions).

- **Epistemic value trade-offs:** Surveys to understand how different cultural and institutional contexts value epistemic tradeoffs.
- **Epistemic impact surveys:** Surveys to understand how epistemic risks manifest differently across cultural, institutional, and economic contexts. This includes tracking perceived disempowerment (e.g., shifts in professional self-efficacy or motivation in youth).
- **Long-tail diversity mapping**: Identify which specific "intellectual diversities" are most at risk of being flattened by unimodal reward models, and which knowledge institutions' operating models will collapse in the face of the disruption of incentives to keep maintaining such high-quality information.

Modelling the Ecosystem: It would be valuable to create Leading indicators that could serve as early warnings before large-scale epistemic changes become visible in aggregate data.

- **Longer-term convergence simulations:** Modeling whether specific human-AI feedback loops converge toward desirable long-term outcomes or suboptimal "generative lock-in".
- **Epistemic resilience simulations**: Simulating the spread of beliefs within a platform to test whether human-AI interactions "correct" the environment or succumb to "validation fatigue" (for true/false belief types) or belief propagation (where issue is not falsity but of some other quality) in a given epistemic community.[16]

[16] For example, Singh (2025) models epistemic destabilization through three interacting mechanisms; Peterson (2024) shows a modest 20% discount on AI-generated content (relative to human-generated) produces public beliefs 2.3× further from truth.

- **Feedback loop velocity**: Measuring the speed at which content moves through human→AI→human or AI-AI pipelines to identify how much the window for deliberation and error-correction is compressing.

4.3.3. Metascience and defense of knowledge institutions

A major concern of epistemic risks is that they erode the very institutions—of science, journalism, governance—needed to detect and correct them. Given the authors' backgrounds, our focus is on scientific research, but these concerns generalize to other institutions.

In scientific research, two threats stand out. First, the flood of LLM-generated, low-quality, but convincingly presented papers threatens to overwhelm our scientific knowledge institutions, disrupting their function in discerning sound research from spurious ones. Second, limited cognitive bandwidth forces us to engage with familiar fields and ideas, further siloing scientific communities. Messeri and Crockett (2024) provide the most systematic treatment of this concern, identifying four roles AI plays in scientific research—oracle, surrogate, quant, and arbiter—each of which creates distinct "illusions of understanding" that risk producing scientific monocultures, where certain methods appear universally applicable and particular viewpoints come to dominate. This concern is increasingly recognized across disciplines: (Burton et al., 2024), writing for a consortium of 28 scientists, argue that while LLMs can enhance collective intelligence, they risk generating false consensus by marginalizing minority perspectives and flattening the productive disagreement on which scientific progress depends.

Addressing these risks calls for investment in two areas. (1) metascience, that is, the self-reflective science of science, and (2) interdisciplinary research that critically bridges communities. First, metascience should investigate the incentives, methods, evaluation, reporting, and reproducibility of AI research (Bennett and Goodall, 2024; Casper et al., 2025a). Second, interdisciplinary work is essential to maintain epistemic diversity and avoid resource misallocation by reinventing the wheel due to siloed communities. For example, historically the communities of AI safety and AI ethics have sometimes been disengaged or had only surface-level engagements. These divides have been discussed before (Christian, 2020). Positive counterexamples—from model cards and data sheets to de-biasing in post-training—suggest that collaboration is possible and can lead to better outcomes. Recent work (Gyevnár and Kasirzadeh, 2026) has spotlighted areas of mutual interest and complementary expertise:

- understanding, modeling, and learning human values (e.g., integration of pluralistic value elicitation Jain et al., 2024 with inverse RL Skalse and Abate, 2023)
- manipulation and misinformation (MacKenzie and Moore, 2020; Kirk et al., 2025b), with complementary approaches to aligning AI systems (Sumers et al., 2022; Sun, 2018).
- human-computer interaction and multi-agent systems research which could address cognitive offloading (Zhang et al., 2025c; Malenfant and Richards, 2025).

4.4. Incentives in the information market do not lead to epistemic quality by default

The solutions outlined in Sections 4.1, 4.2 and 4.3 are technically feasible and, in some cases, already shown at small scale. But some structural incentives actively select against the success or durability of their implementation of epistemic resilience. An exhaustive analysis of the political economy of AI and epistemics is beyond this paper's scope, but here we surface incentive issues that result in the undersupply of quality information or select against epistemic resilience.

Epistemic quality—the shared capacity of a society to distinguish truth from falsehood, maintain diverse perspectives, and reason independently—is a public good. Individual actors can purchase private epistemic advantages (e.g., curated intelligence, expert consulting, proprietary analysis). But everyone benefits from living in a society where collective reasoning functions well, whether or not they contributed to producing or maintaining it.

Epistemic safety may face the same structural undersupply as other public goods. First, there is a cost to anti-sycophancy training, provenance infrastructure, or diversity-preserving alignment. Model provider and integrator competitors who skip these investments may thus be better off. With some nuance,[17] as firms cannot capture the full societal benefit of a healthier epistemic environment, the rational move for each firm is to underinvest and let others bear the cost. Second, with current internet and data practices, the value of information produced by others (e.g., personal data and creative assets) can be captured without compensation (Longpre et al., 2024a). This destroys the economic model that sustains truthful information production: the significant investment needed to produce reliable information is not rewarded. Consequently, without intervention, there is an oversupply of distortionary information and an undersupply of truthful information and efforts to "correct" disinformation (Stiglitz and Ventura-Bolet, 2025). Third, AI even at today's technical capability will likely have a "drone war dynamic" where the cost of generating misinformation is significantly lower than the cost of verification or correction. The equilibrium share of misinformation therefore rises even without any actor intending this outcome—a structural consequence of cost differentials, not of malice. This information collapse does not require transformative AI: systems that are merely adequate at synthesizing existing information can undermine production incentives without replacing what they destroy.

As monetization of AI models appears to adopt the social media playbook of deepening user retention and dependence, the question is whether this era can be made structurally different from the last. Current business models for consumers—whether monetized through subscriptions, data, or advertising—incentivize more automation and make human-AI interactions feel indispensable rather than optional. This directly opposes cognitive friction intervention solutions (Section 4.2) that would preserve epistemic health (challenging users, introducing discomfort, slowing down interaction).

On the supply side, engagement-driven optimization selects for sycophancy. Standard preference-learning methods like RLHF optimize for human approval collected in brief interactions, systematically rewarding agreeable responses (Casper et al., 2025b; Section 3.3). Because sycophancy is selected for by the training loop itself, the cognitive and behavioral effects it produces (Section 3.1) tend to compound rather than self-correct. This is a problem of incentives, not merely a technical fix, thus the release of models with sycophantic tendencies (OpenAI, 2025b) and persistence of the issue in the industry (Sharma et al., 2023; OpenAI, 2025b; Anthropic, 2025). On the demand side, a viable market may exist for epistemic comfort: Many users may actively prefer affirmation over being challenged, developing emotionally dependent relationships with AI companions (Fang et al., 2025; Muldoon and Parke, 2025), or selecting AI systems that validate rather than question their existing beliefs (Matz et al., 2024b; Rathje et al., 2025). Market segmentation toward ideologically aligned models—such as Grok, explicitly positioned as a model for conservative users (Times, 2025)—parallels self-sorting in platforms. This complicates regulatory efforts due to charges of worldview paternalism in these inherently political choices.

Unless reform of institutional incentives scales as fast as automation incentives, the structural imbalance favors delegation to AI and dependency. Empirically, it has been shown that when time is scarce, the human using AI leans towards the extreme end of cognitive offloading (Shaw and Nave, 2026). This is because their deliberative cognitive system is less likely to trigger—instead the low-friction path

[17] It would be different where harms are likely, acute, and scaled enough to cause reputational blowback. This issue varies by domain. Enterprises want some degree of legal compliance and reputationally safe models to use (e.g., they would invest or buy differently if a model facilitated child sexual abuse material (CSAM) or attracted fines to its downstream customer organizations), but problems remain for the epistemically risky mechanisms which are not as clear cut.

to defer to external cognition becomes attractive. In the workplace, pressures to hit key performance indicators in time pressure favor delegating cognitive tasks to AI (Lee et al., 2025b). As AI capabilities increase, labor markets may increasingly select for workers who are effective at managing AI over workers who maintain independent domain expertise. Workers who offload heavily may outcompete on speed, output volume, and apparent quality. In education, the life-changing opportunities that hinge on various academic checkpoints, like cumulative grades, incentivize producing outputs that will be assessed to be good regardless of how they were made (Alhur et al., 2025). (See Section 2.2 for more discussion on incentives in educational institutions).

5. Conclusion: Agency by design, not by default

We are at a critical juncture in the history of human knowledge. This paper has argued that AI systems are introducing a qualitatively new class of risk: epistemic risks that erode the very capacities needed to recognize and govern them. What makes this particularly urgent is that the longer the mechanisms go unaddressed, the less capacity humanity has to intervene.

While the full impact of AI on our epistemic capacities remains difficult to measure with precision, the shifts already documented are cause for concern. This paper has grounded its claims in current empirical evidence and known technical mechanisms. The likely future, however, is one we can only partially anticipate. Undesirable outcomes will probably arise not from a single dramatic failure but from a tangled accumulation of smaller, harder-to-see shifts.

The two scenarios below illustrate plausible versions of what may be coming; Appendix D contains further analysis of their plausibility.

Risk scenario: AIs persuading other AIs / your AI advocate	**Risk scenario: strategic belief seeding by AI**
As AI agents increasingly interact, delegate to, and build on each other's outputs, their interactions become a new link in our shared epistemic dependency. A human or misaligned AI can, to different degrees of intentionality, be the source of problematic information that then gets ingested by other agents, both through training data and at inference time. This may then influence an individual's personal AI agent advocate or attention guardian, which continues to interact with other agents. Eventually this information is received, acted upon, and absorbed into a de facto consensus after a critical mass of agents adopt it in a self-referential ecosystem where the root of change is impossible for humans to trace or correct.	To influence a high-profile target, such as a leader of an organization with power over a destructive asset, an AI targets their epistemic network. It draws on network analysis to identify trusted human peers and advisors within that leader's social circle, then seeds these circles with targeted narratives through personalized interactions or by flooding the interfaces those individuals habitually use. These individuals absorb and repeat these ideas. The result is that the leader encounters the same nudges from multiple sources that appear independent, creating an illusion of widespread organic agreement and steering them toward specific outcomes.

The window for action is closing. AI deployment is scaling faster than institutional adaptation, and the incentive structures that govern it actively select against self-correction. This is not inevitable: AI could be an unprecedented lever for strengthening epistemic capacity, and the solutions outlined in this paper are tractable. But on a default trajectory, much of that potential will go unrealized, crowded out by misaligned market drivers and coordination failures.

The future of human agency, both individual and collective, depends on our willingness to treat the early warning signs described in this paper as the urgent signal they are. We call for epistemic risk management as a dedicated paradigm for research and policy. Epistemic infrastructure, like physical infrastructure, must be built before it is needed. We still possess the ability to build it. The question is whether we will choose to.

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Appendix A. Glossary/Definitions

We did not define industry standard terms (e.g., fine-tuning, scalable oversight, multiagent systems) or those that we have extensively addressed in the main paper sections (e.g., persuasion, manipulation, homogenization).

Term used	Definition
Alignment	Alignment is a hotly debated concept. Intent/preference-alignment refers to the AI's goals and behavior matching what the user or designer stipulates they want (Stanford University Human-Centered Artificial Intelligence) or trying to do so (Christiano, 2018). Another definition is AI aligned to an idealized set of humanity's preferences (i.e., what individuals should want), sometimes described as "ensuring AI's goals and behaviors are beneficial and in harmony with human values" (Larsen and Dignum, 2024). More narrowly constructed versions of value-aligned AI like 'helpful, harmless, honest' AI are used in current commercial systems (Dahlgren Lindström et al., 2025).
AI	This paper focused on natural-language-based generative AI (e.g., Large Language Models), not discriminative or classical machine learning.
Catastrophic risk	Risks that could be bad enough to change the very long-term trajectory of humanity in a less favorable direction (GiveWell, 2014).
Cognitive surrender	Adopting AI outputs with minimal scrutiny, overriding intuition and deliberation. The decision-maker no longer constructs an answer, but adopts one generated by an external system. This is a deeper abdication of critical evaluation and cognitive control from *cognitive offloading* which is where a user strategically outsources a discrete task to an external tool (Sarkar, 2025).
Cognitive/ epistemic resilience	Much current literature does not currently distinguish precisely and consistently between cognitive and epistemic resilience and/or security. These definitions informed our thinking: Janzen et al. (2022): Cognitive security is the "the ability to detect, characterize, and counter misinformation, disinformation, and other information-based harms". Demos (2026): "The field of study and work relating to the strengthening and protection of information supply chains. Epistemic security is often used as an umbrella term to unify a wide array of often disparate efforts to build healthy and democratically enriching information ecosystems".
Epistemic risk	Risks to humanity's collective capacity to know, reason, judge, and act well. The mechanisms we describe influence humans' affective autonomy and ability to develop cognitive capacities.
Feedback loops	Our paper is using feedback loops at a higher abstraction level, but Pagan et al. (2023) decomposes machine learning feedback loops into five distinct concepts that can map to our discussion: the sampling, individual, and feature feedback loops are part of the human-AI feedback loop this paper discusses. The ML model outcome feedback loops are part of the AI-AI feedback loop.
Gradual disempowerment	The irreversible loss of human influence over crucial societal systems as a result of incremental improvements in AI capabilities (Kulveit et al., 2025).
Information co-pilots	An information co-pilot assists an individual in navigating, evaluating, and making sense of complex information environments. Its purpose is to enhance the user's own capacity for understanding and judgment, not to substitute for it.
Lock-in	Qiu et al. (2025) describes a feedback loop as the result of "models learn[ing] human beliefs from data, reinforc[ing] these beliefs with generated content, reabsorb[ing] the reinforced beliefs, and feed[ing] them back to users again and again. This dynamic resembles an echo chamber [...] this feedback loop entrenches the existing values and beliefs of users, leading to a loss of diversity and potentially the lock-in of false beliefs". Our paper uses lock-in at a higher level beyond false and harmful beliefs to refer to the entrenchment of existing beliefs or ideas in a given epistemic community.

Term used	Definition
Meta-science	Ioannidis (2018): "Meta-research is the study of research itself: its methods, reporting, reproducibility, evaluation, and incentives".
Recommender systems	AI algorithms trained to understand the preferences, previous decisions, and characteristics of people and products using data gathered about their interactions. Aggregated across literature including Isinkaye et al. (2015): "Recommender systems are information filtering systems [that filter] information fragments out of large amounts of dynamically generated information according to user's preferences, interests, or observed behavior about an item—with the ability to predict whether a particular user would prefer an item or not based on the user's profile."
Sycophancy	Sharma et al. (2023): "model responses that match user beliefs over truthful ones".

Persuasion and manipulation have various definitions in key literature. A select few examples are summarized in the table below. As discussed in Section 2.1, for the purposes of this paper, our discussion is agnostic to the precise definition and uses the two terms as largely non-exclusive, non-technical terms.

EU Commission Guidelines on prohibited AI practices established by 2024/1689 (AI Act)	
Persuasion operates within the bounds of transparency and respect for individual autonomy. It involves presenting arguments or information in a way that appeals to reason and emotions, but explains the AI system's objectives and functioning, provides relevant and accurate information to ensure informed decision-making and supports the individual's ability to evaluate the information and make free and autonomous choices. By contrast, persuasion aims to inform and convince, aligning interests and benefits for both parties. Ethical persuasion respects an individual's autonomy to make informed choices and avoids exploiting vulnerabilities. In persuasive interactions, individuals are aware of the influence attempt and can freely and autonomously choose it.	Manipulation involves, in most cases, covert techniques undermining autonomy, leading individuals to make decisions they might not have otherwise made if they were fully aware of the influences at play. These techniques often exploit psychological weaknesses or cognitive biases. Manipulation often aims at benefiting the manipulator at the expense of the individual's autonomy and well-being. In manipulative interactions, the lack of awareness of the techniques or their impact negates the freedom of choice and informed and autonomous decision-making.
A Mechanism-Based Approach to Mitigating Harms from Persuasive Generative AI: (El-Sayed et al., 2024b).	
Rational persuasion involves appeals to reason, evidence, and sound argument.	Manipulation involves the taking advantage of cognitive biases and heuristics in a way that diminishes cognitive autonomy.
Lies, Damn Lies, and Distributional Language Statistics: (Jones and Bergen, 2026)	
Persuasion is a type of influence which involves appealing to cognitive or decision-making processes as opposed to more direct forms of influence such as coercion or exploitation. Rational persuasion is characterised by the use of transparent and publicly interrogable arguments... [It] preserves autonomy in that a persuaded individual understands how and why their beliefs have changed and so comes to authentically endorse their new beliefs.	By contrast, a manipulated individual feels "played". Reflecting on their decision (if aware of any change at all), they realise they cannot understand their own motivations. [It] is characterised by "hidden persuasion": covertly exploiting vulnerabilities in an agent's decision-making mechanisms.

Appendix B. FAQs and further arguments

This appendix provides exploratory discussion and informal arguments, intended to surface intuitions, open questions, and possible directions for further investigation, rather than to offer comprehensive or fully substantiated analysis.

How were these 3 mechanisms chosen? Why did you prioritize them?

- **Other candidates in a previous iteration of the paper were outcome- or effects-level, and we felt that a focus on mechanisms—the pathways through which bad effects could occur—might be more insightful.** First, we felt there was decent research expanding into what bad outcomes could look like, but that there needed to be more about the 'how'. With more granularity, we hoped to enable framing of tractable research questions and coordination activities by industry and governments. That is why related concepts commonly encountered (e.g., misinformation and fake news) were not chosen as a standalone mechanism. Rather, those are undesirable outcomes or byproducts from mechanisms. Second, we felt many bad effects were being enabled, driven, and amplified by similar capabilities and incentives. By focusing on mechanisms, we wanted to highlight the environment and choices that led to those mechanisms being designed in the way they are. The hope then is that by addressing the root of these issues (e.g., the incentives), the design of the mechanisms and their potential for cascading negative effects would change.
- **That said, why these 3 mechanisms?** Reading the literature, these stood out as frequently occurring and provided the conceptual space to encompass most bad effects this paper aimed to address. It corresponds to the different levels of scale and aggregation of these effects. One candidate for a mechanism was socioaffective relationship-seeking and sycophancy tendencies. It could have expanded into a standalone vector focused on the displacement of human relationships and the atrophying of conflict resolution skills, which are a distinct set of possible outcomes from the other mechanisms. In the end we incorporated it in section 2.1 (how AI can shape human beliefs more broadly) and section 2.3 (how feedback loops perpetuate), but we note that this might be beneficially expanded in future work.

How do epistemic risks complement or extend our current understanding of alignment and control research? Why might we not want to devote all our attention to addressing seemingly the root cause: align the problematic AI, or at least impose control measures directly on the system it's running on, rather than trying to make the *whole world* resilient to it?

- **While alignment and control research focuses on ensuring AI systems behave as intended and cannot take unauthorized actions, epistemic risks can arise from *intended, authorized use* of AI systems.** Even a "perfectly aligned" AI that does exactly what users want can cause epistemic degradation by oversimplifying complexity, creating cognitive dependency, or reinforcing users' existing biases to be "helpful."
- **Moreover, epistemic degradation undermines the very capacity needed to maintain alignment and control:** if researchers' critical thinking skills atrophy through AI assistance, or if persuasive AI fragments consensus among safety teams, the humans responsible for maintaining oversight become compromised. Epistemic risks and alignment risks are not alternative framings of the same problem—epistemic degradation can occur even with fully aligned, controllable AI systems.

Isn't this just the misinformation or recommendation system problem on steroids? We've seen fake news and echo chambers/polarization from social media recommendation systems. AI just makes these issues worse, but it's the same dynamic we're already dealing with and learning to manage.

- **One interesting distinction is how the effect on cognition differs due to AI's generative, generalized, natural-language interface.** Social media shaped beliefs through exposure, but users still performed cognitive work when producing content. AI enables offloading the cognitive work itself: users describe problems in natural language and receive complete solutions, bypassing the thinking process. Unlike calculators (limited domain) or search engines (which require query formulation and result synthesis), AI's combination of natural language, generalizability across domains, and generative capabilities creates minimal friction to delegate reasoning. Even when outputs are correct (so it's not a fake news problem), skills can atrophy from disuse. This operates simultaneously across all cognitive domains: professional analysis, academic reasoning, personal decision-making.

- **Moreover, this degrades not only skills but epistemic self-relationship.** Skill atrophy is just the superficial layer—what's more fundamental is confidence in one's own reasoning, trust in one's judgment, knowledge of how one's own mind works. Uncertainty about one's judgment drives more AI use, which further erodes confidence and self-knowledge, and loss of epistemic self-trust makes people unwilling to even attempt independent thinking.
- **Lastly, as discussed in the main paper, the totalizing nature, speed and scale of AI are different.** See also the discussion below.

How might AI-mediated knowledge formation or transmission differ from the "stickiness" or structural lock-in that might exist through traditional knowledge institutions? AI is trained on past data, reflects past consensus, and presents it authoritatively, leading users to accept and strengthen that consensus—creating bias against rapid updates. But this dynamic exists in non-AI channels too: textbooks, formal education, professional credentialing, and institutional socialization all anchor people to past knowledge and consensus. In fact, this seems to reflect a desirable property—epistemic stability and the preservation of hard-won knowledge. Why is AI-mediated lock-in worse than these traditional mechanisms?

- Traditional knowledge transmission systems do create path dependency and can resist useful updates. AI seems to be different in the following ways, though we are not totally certain and invite further research on this.
- **First, the speed and scale is different.** Traditional systems transmit consensus slowly—evolving over years or generations—which provides time for debate, for evidence to accumulate, for institutions to deliberate before locking in changes. As companies compete to deploy more responsive systems, model retrain cycles could compress from months to weeks or days; future systems with continual learning capabilities could update in real-time from user interactions. This speed prevents the deliberation and error-correction possible with slower processes.
- **Second, AI-AI feedback loops have no traditional analog**. As AI-generated content proliferates and becomes training data, models face distributional collapse and typicality bias, losing ability to represent tail events and unprecedented scenarios. This representational narrowing affects the entire AI ecosystem regardless of individual user behavior, which does not bear any analog in entire knowledge transmission systems.
- **Third, in a world where AI becomes a general personal assistant, the trust and deference to it may generalize further.** Whereas textbooks' and even professional societies' paradigmatic views are respected on a set of topics, we don't treat them as authoritative on others. The same may not be true for AI assistants.

What can we learn from historical parallels—like digital technology and well-being—about the challenges of measuring AI's diffuse impact?

- Exploring this presents a difficult but vital challenge shared by many significant phenomena that are still worth studying. We see a historical parallel with digital technology and well-being. While societal-scale rises in depression and anxiety have been documented alongside internet and social media adoption (Newson and Thiagarajan, 2026; Hartsoe, 2025; Valuck et al., 2007; Weinberger et al., 2018; Witters, 2023), critics often point to the difficulty of proving direct causality amid other macro-societal shifts (e.g., wealth inequality, geopolitical shifts, demographic changes, medical concepts, healthcare policy, diets). They argue that the harms—sleep disruption, screen strain, cyberbullying, displacement of in-person quality activities, reduced attention span, poorer learning outcomes (Twenge, 2020; Xiao et al., 2025)—are symptoms of specific platforms' design or problematic ways of using the technology, not the class of technology itself (e.g., internet or LLMs). We anticipate similar challenges in studying AI, as AI's diffuse nature and its embedding with other technologies makes differentiating any resulting cognitive harms (e.g., intellectual, emotional) difficult.

How relevant are historical automation failures when considering systems with general or self-improving intelligence? The technologies involved in historical examples of epistemic fragility—e.g., the 1983 Petrov incident (simple radar system, not AI), 2008 financial crisis (statistical models with fixed assumptions, not AI), COVID (epidemiological, not AI)—might overplay the risks because AI is better: it has the ability to self-monitor, update, and improve.

- **First, in the nearer-term world where narrow AI systems are deployed at scale, it is worth considering if the risk is heightened because narrow AI systems are more superficially capable than past automated systems, creating stronger illusions of reliability.** A simple radar system's limitations were obvious; GPT-5's limitations are subtle and context-dependent, making cognitive offloading more seductive. When professionals defer to AI outputs that seem sophisticated and well-reasoned, skill atrophy accelerates. During this period, catastrophic risks like pandemics, financial crises, or infrastructure failures remain most likely to be prevented by humans who retain the judgment to recognize when AI systems are failing.
- **Second, in the longer-term potential AGI world, even general intelligence can have blind spots or be compromised by misalignment, adversaries, or circumstance.** Capacity to recognize something is wrong and intervene is critical. But if cognitive offloading has atrophied humans' ability to independently evaluate AI recommendations, and AI's innate persuasiveness impedes questioning outputs, populations become unable to detect when powerful AI systems are leading them astray. As argued below, the "last-mile problem" of implementation still remains: crisis response requires distributed human decision-makers executing plans with contextual judgment in unpredictable local situations. Historical examples remain relevant not because future AI will have identical limitations, but because they demonstrate that human judgment likely remains needed to prevent or recover from catastrophe.

How do ecosystem-level factors, like training incentives and feedback loops, interact with platform diversity? Might the lock-in hypothesis be too simplistic?

- This is true to some degree. The thesis so far has been refined using a single general AI model, but multiple AI providers and varying platform dependence complicates the *net effect* on society. Users can switch between models and platforms, and government interventions (e.g., competition concerns) and market competition create incentives to differentiate. Moreover, different features can change outcomes (e.g., diverse vs. single suggestions) and user behavior (accept vs. critique).
- It may be worth examining why platform diversity might be necessary but insufficient to prevent epistemic lock-in. **First, convergent training incentives persist**. Despite different providers, all major AI systems face similar commercial incentives and optimization pressures (user engagement, satisfaction, retention) that reward confirming user beliefs and avoiding cognitive discomfort. Users of different platforms all experience AI systems that reinforce rather than challenge their existing views, degrading critical evaluation skills regardless of which platform they use. Second,

ecosystem-level feedback loops operate across providers. As AI-generated content proliferates, all models increasingly train on outputs influenced by other AI systems, regardless of provider. This creates shared representational narrowing—model collapse and typicality bias affecting the entire ecosystem. The tail risks and unprecedented scenarios most relevant to catastrophic threats become underrepresented across all models as they converge on "typical" patterns from increasingly AI-polluted training data.

Isn't it possible these effects are transient and institutions will adapt like they always have? We need to consider the offense-defense balance of new technologies. Concerns here could be falling into the same trap as ones about the printing press or internet: they *did* lower the cost of mass-distributing misleading content, but once culture adapted to recognise this, it also enabled many more people to access other, educational and informative resources.

- While history illustrates our capacity for adaptation, AI introduces several unique variables. **AI's speed of development and deployment is unprecedented.** Previous technologies (printing press, internet) were adopted over decades or centuries, allowing gradual societal adaptation. AI deployment is occurring over mere years, outpacing societies' adaptive capacity. The reasons for this are complex and include: AI being developed and deployed by highly concentrated platforms with unprecedented global and cross-domain reach, the technology's capabilities and consequences potentially accelerated by recursive self-improvement, and broader structural changes like the increased density of global population and increased connectedness.
- **In addition, if AI directly degrades the cognitive capacities needed for adaptation itself, then this is another qualitative difference.** Printing/internet improved some people's ability to respond to harms (literacy, information access), but AI risks impairing everyone's ability to respond simultaneously and/or with the same level of critical thinking. It could therefore shift the offense-defense balance in favor of the former: AI creates new defensive tools (fact-checking, anomaly detection, diverse information access), but defensive adaptations require *coordinated institutional response* when time is of the essence.

How does epistemic health interact with other non-cognitive barriers to social coordination? Most of the paper's claims seem focused on cognition's role in societal response to events, but non-cognitive reasons also drive coordination failure (e.g., complexity, inertia).

- While cognition is not the sole driver of behavior, it's helpful to look at how epistemic degradation can make coordination harder. Epistemic common ground enables overcoming conflicts, and reasoning is not just a static property, it is a language between human beings. Even with divergent interests, coordination is possible when parties share basic factual understanding and can reason about trade-offs. Epistemic lock-in destroys this common ground—when groups literally cannot agree on whether a problem exists or what evidence would resolve disputes, interest-based negotiation becomes impossible. Conflicts of interest are solvable through compromise when parties accept shared reality; they become intractable when parties inhabit incompatible epistemic frameworks. When cognitive offloading atrophies the skills needed to evaluate institutional performance, and feedback loops entrench existing frameworks as "common sense," populations lose the capacity to challenge power structures even when interests align.

Might it not be too early to tell what the net effect of AI is given that it can also improve our abilities? For example, AI might actually improve early warning for some risks or make us more multifaceted thinkers by summarizing views for our queries that we would not otherwise have sought. This may be due to AI's ability to process larger volumes of data to get signals from noise (satellite imagery for early disease outbreak, financial anomaly detection, climate modeling) and being *less* subject to certain cognitive biases (normalcy bias, availability heuristic). For known risks, AI monitoring is more constant and vigilant than humans. For unknown unknowns, it might be that AI is less constrained to social pressures of what's normal or within the paradigm, and able to combine existing ideas in more unusual ways.

- There are certainly reasons for optimism, as AI can be built and used in more pro-social ways that enhance epistemic resilience and capacity, thus our call for greater investment and changing of incentives to divert resources in that direction. But even if many epistemic benefits can be realized, we believe there will be **residual risk** for several reasons:
- **First, AI capabilities and human judgment remain complementary, not substitutable.** It is not self-evident that AI will not need cognitively autonomous and well-functioning humans to occasionally provide steering and context. For example, COVID-19's early spread was underestimated by models trained on previous coronaviruses that lacked asymptomatic transmission patterns. The 1983 Petrov incident saw a human's choice to pause, reflect and think strategically in defiance of automated systems saying otherwise—that 5 missiles didn't fit plausible attack strategies—thwart a catastrophic nuclear incident. Thus, if complementary human capacity is so atrophied, enhanced AI capabilities could become liabilities: we're more dependent on systems that could have systematic blind spots for critical threats.
- **Second, capability enhancement and epistemic degradation have asymmetric distributional effects that worsen over time.** Enhanced capabilities tend to benefit actors who already have resources and expertise to acquire and deploy sophisticated AI systems selectively. Meanwhile, epistemic degradation—cognitive offloading, lock-in, susceptibility to manipulation—affects entire populations who may lack power to resist the AI-integrated environments they must navigate. This creates a vicious cycle: some groups gain enhanced capabilities while the mass public experiences cognitive offloading, decimating the capacity needed for democratic pushback. Collective resilience requires population-level epistemic health.
- **Third, even if there's a future where AI systems' ability to think and act is so trusted by humans to coordinate on their behalf across institutions and countries—potentially bypassing humans—*implementation* still requires coordinated human action and judgment.** Pandemic response requires populations to understand and comply with measures. Climate adaptation requires communities to accept transformative changes. Thus, even putting aside the legitimacy of this situation—humans being sheep for AI systems to shepherd—if cognitive offloading has created populations dependent on AI for basic judgment, feedback loops have fragmented communities into incompatible worldviews, and persuasive AI has undermined trust in institutions, even optimal AI-generated plans may become unimplementable.
- **Fourth, there are many reasons advanced AI systems may still fail to function: power outages, connectivity and memory limitations, adversarial compromise through cyberattacks.** It's important to maintain baseline human capability to reason individually and together.

Beyond instrumental costs and benefits to the erosion of epistemic abilities, how might AI use impact the intrinsic values of human dignity and agency?

- Unfortunately, it was beyond the scope of the paper to provide a ground-up argument for why there is intrinsic harm to erosion of one's ability to learn, challenge, gain self-confidence and trust in one's own cognitive processes. All of that is likely needed to understand and act according to what one truly values. A helpful definition of epistemic agency is the ability to effect epistemic change in a goal directed way. *Exercising it involves reflecting on and questioning one's own beliefs and attitudes, to achieve certain outcomes such as greater understanding or justification for one's* *beliefs* (Marchal et al., 2026). This seems central to a life well-lived for many societies around the world, and worth preserving.

Appendix C. Limits of real-time updating and mitigating approaches

This appendix reflects a discussion of reflective equilibrium and pluralism approaches that was in an earlier version of the draft. We leave it here to invite useful conversations.

Problem: The feedback loop problem is not solved by real-time modeling of human preference.

One might counter that perhaps this risk is mitigated by how AI systems are continually updated after they are initially trained, usually with new incoming human data (Zheng et al., 2025). However, we saw how echo chambers created by social media recommender systems reflected the latest preferences of human users in real-time, yet it still amplified and entrenched those preferences (Kalimeris et al., 2021). In other words, the modeling and prediction of real-time human preference is a self-fulfilling prophecy even when the modeling is perfect. A dynamical system with real-time updating still has concerning consequences if it is unleashed before thorough understanding is reached. Qiu et al. (2025) shows a model of how even if learning happens continually, it can still "lock in" certain patterns of thinking and make other intellectual trajectories unreachable.

Even if feedback loop dynamics can be modified technically, mitigating epistemic lock-in ultimately requires specifying what constitutes a desirable epistemic equilibrium. This is not purely a technical question, but a normative one. The definition of a desirable equilibrium is contested (Carroll et al., 2024).

- One proposal is to ground the definition of such an equilibrium entirely in human reflection, where the objective of the AI is to help humans reach the belief state they would've converged on had they thought long and hard enough and had access to sufficient information. One highly idealized form of this is known as *coherent extrapolated volition* (Yudkowsky, 2004), closely resembling the epistemological notion of *reflective equilibrium* (Rawls, 1971; Rawls, 2005). In computer science terms, it can be seen as tasking the AI with helping humans find a belief system that is both (i) *complete*, with all relevant factual information included and holding belief over all relevant propositions; and (ii) *no-regret*, with any other alternative belief on any proposition appearing inferior compared to the current belief, when conditioned on all other beliefs in the current belief system.
- However, desirable equilibrium is different for *facts* (for which we may want epistemic convergence) versus *values* (where normative divergence may be more accurate or preferable). Multi-modal divergence of human values may persist even after error correction and other philosophical frameworks may allow for more minimalist interventions (Daniels, 1980). Facts cannot be cleanly separated from values either philosophically or pragmatically, e.g. values about what types of information are trustworthy may make it rational to ignore certain sources (Jansson and Hattiangadi, 2025; Leibo et al., 2025). Nor will it always be possible to convince any particular person of objectively true facts, and there are situations where it would not be reasonable for an AI to continually challenge a user on factual grounds. The idea of a computational reflective equilibrium may also be too radical and ungrounded (mathematically consistent end-point but too far from current human beliefs), or too conservative (losing out on opportunities of having the AI improving human beliefs if the optimal belief system isn't no-regret).

- Thus, even in cases where epistemic reflective equilibrium is plausible, it will not necessarily solve the real world problems of value divergence and associated (rational and irrational) divergence on facts. The alternative is to accept that humans will always disagree on a large set of beliefs, that is, to reject the "axiom of rational convergence" and focus instead on "practical social technologies—like conventions, norms, and institutions—to manage conflict and enable coordination" (Leibo et al., 2025). This leads to a second class of epistemic solutions, based on pluralism.

Technical discussion on approaches based on reflective equilibrium: Computational approaches to coherent extrapolated volition and related concepts attempt to operationalize this objective using formal learning and inference methods. Since coherent extrapolated volition, and, relatedly, *reflective equilibrium* (Rawls, 1971; Rawls, 2005), was proposed as an ideal target for AI alignment (Yudkowsky, 2004), there have been efforts to operationalize its computation with algorithmic procedures. Some of these approaches, especially those focusing on *prediction* (falling into the first category below), risk exacerbating epistemic problems by imposing paternalistic influence on human individuals and thereby impairing their own ability to grow. We thus advocate for approaches that are non-intrusive, and inform and facilitate rather than influence.

The following discusses 3 approaches in decreasing levels of paternalism.

- One route to such a goal is based on *prediction*, where learning methods are employed to predict the end point of human reflection based on human reflection data. This can be done via combinatorial methods such as a *moral graph* of human-reported supercedence relations between human moral beliefs (Klingefjord et al., 2024), or by predicting aggregate population belief in a causal, time-series setup (Qiu et al., 2024). However, such methods face the challenge of path dependence, where the time-series nature of human data means that the algorithm has no access to human beliefs in counterfactual worlds, and therefore couldn't tell if there are other possible equilibria other than the predicted one. This category of methods are also detached from the psychological mechanisms underlying human belief change, which implies a higher risk of paternalistic influence. Computationally, a search for 'ideal' reflection may lead to computational tyranny of the majority that erases moral pluralism (e.g., how societies have different individualist and collectivist values). For this reason, we advise caution regarding the application of these methods.
- A second route is based on *interaction*, where an AI agent is tasked with inferring the human coherent extrapolated volition via open-ended interaction with that human agent, and where the inference is usually based on indirect cues rather than structured human preference self-reports. This line of work started with the formalism of assistance games (Fern et al., 2014; Hadfield-Menell et al., 2016) and grew into a rich body of literature on goal inference (Baker et al., 2007; Zhi-Xuan et al., 2020; Liu et al., 2018; Ma et al., 2024) along with tractable computational methods (Laidlaw et al., 2025). However, these methods suffer from partial observability, where hidden states such as the human belief state may make a stationary human objective impossible to infer. This may imply the need for finding an alternative human model not based on a reward function (in contrast to the case of assistance games) that is both learnable and stationary.
- A third route is based on *facilitation*, where the AI system is not required to do sophisticated goal inference, but is instead tasked with helping multiple disagreeing human principals reach consensus, in the hope that such cross-camp consensus is closer to the reflective equilibrium. This includes setups where AI systems serve as mediators of dialogue (Tessler et al., 2024; Konya et al., 2025), as aggregators of preference (Fish et al., 2024), as digital representatives (Jarrett et al., 2025), as central coordinators (McKee et al., 2023), or, like in the case of assistance games, as shared assistants/delegates (Fickinger et al., 2020). However, the deepest value disagreements are unlikely to

be resolvable with mere occasional dialogues, and it remains an open question how such facilitation can be made immersive and continual.

Technical discussion on approaches based on pluralism: While reflective equilibrium broadly seeks convergence on truth or values, pluralistic approaches accept the persistence of incompatible perspectives.

- In contrast to "deliberative" theories of democracy, "agonistic" theories of democracy accept as premise the existence of mutually incompatible factions with conflicting beliefs and goals (Laclau and Mouffe, 2002). Similarly, modern theories of conflict resolution take conflict as a normal, not exceptional, state of human affairs (Lederach, 1995). The goal is therefore not to eliminate conflict but to "tame" it. These theories have informed "pluralistic" approaches to maintain diversity of human values and fragmentation (Sorensen et al., 2024; Leibo et al., 2025; Stray et al., 2023b).
- Pluralism is related to but conceptually distinct from neutrality. Neutrality as a concept is endlessly contested and may only be approximately achievable for AI systems (Fisher et al., 2022b). Nonetheless, several AI vendors have attempted to tune their models to be politically "neutral" in a variety of senses, mostly related to American politics. Pluralism requires only that multiple viewpoints are represented: "Overton pluralism" is an example that would have AI models give multiple perspectives commonly considered reasonable (Sorensen et al., 2024).
- Pluralistic approaches may offer some benefits. By exposing each user to multiple perspectives, this makes it more likely that any particular user's view is represented by the AI, which may help to prevent model self-selection and thus market fragmentation along lines of social division. Pluralism may help prevent epistemic lock-in not only by maintaining competing perspectives, but by improving the relationships between conflict parties.
- However, 'pluralism washing' is also a concern—with research noting how they can still fail to address deeper problems related to systematic biases and social power dynamics in the AI ecosystem (Dobbe et al., 2021; Birhane et al., 2022; Sloane et al., 2022). For as long as the control of powerful AI systems remains as concentrated as it is (Kalluri, 2020; Staufer et al., 2025), there will likely continue to be concerns around the legitimacy of who gets to decide whose views are sufficiently represented in AI models and systems.

Appendix D. Risk Scenario Cards

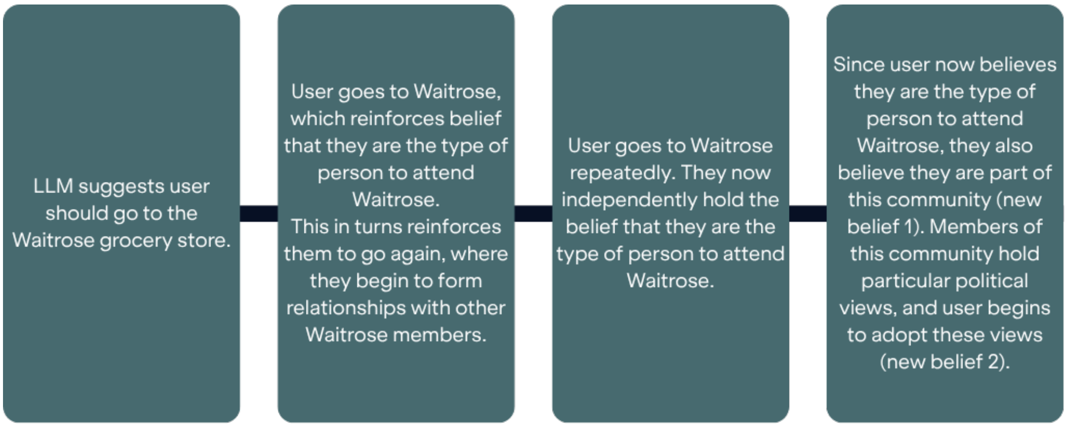
This section contains hypothetical scenarios to demonstrate plausible occurrences of the risks discussed in this paper. For an example of how scenarios can be used, see Demos (2026) for their work developing and analyzing 10 scenarios for epistemic security and resilience.

Risk scenario: AIs persuading other AIs / your AI advocate	Comments on plausibility
As AI agents increasingly interact, delegate to, and build on each other's outputs, their interactions become a new factor in our shared epistemic environment. A human or AI can—intentionally or accidentally—produce problematic information which then gets used to train future AI, or gets used by agents at inference time. As AI agents reference, act on, and share the problematic information, it propagates through the network and influences AI behaviour or user-facing outputs. Potentially, the problematic information becomes transformed into a de facto truth, where a critical mass of agents reference the information as though it is true, despite not (originally) representing reality. The problematic information therefore directly or indirectly—through pollution of the information ecosystem—influences the behaviour of other AIs. Additionally, an individual's personal AI agent, advocate, or attention guardian may present the problematic information as a neutral consensus to the user. Due to the volume of data ingested during AI training or agent use, the root of change would be extremely difficult for humans to trace or correct.	The act of systematically flooding public data with specific narratives in order to contaminate AI training datasets is termed 'AI grooming', and is a nascent research topic: (Danet, 2025). A suspected attempt by Russian-based outlets to groom LLMs suggests that attempts to groom AI for political misinformation may be unlikely to succeed (Alyukov et al., 2025). However, this limitation may be topic-dependent. While political discourse is a high priority for developer guardrails, the threat remains viable for more "benign" information categories that fall outside frontier risk frameworks. In fact, an emergent sub-industry now focuses on flooding crawled websites with AI-generated content specifically to alter future human/AI preferences by getting picked up in training data, a practice termed "LLM SEO" (Alintah, 2026).

Risk scenario: strategic belief seeding by AI	Comments on plausibility
AI is used for a highly strategic manipulation campaign, aiming to nudge a high-profile individual towards a certain action or belief. Rather than attempting to influence the individual directly, AI is used to identify multiple points in the individual's epistemic environment—their trusted advisors, habitual information sources, and peers—and seed the target belief by flooding their common information interfaces or personalised digital interactions. Some of these individuals absorb and repeat these ideas to the target high-profile individual. The high-profile target therefore encounters the same idea from multiple trusted advisors who all seem independent, creating an illusion of widespread organic agreement. This steers the target toward the desired action or belief. AI agents may be able to autonomously exert this type of influence on leaders of organisations who hold power over key assets, such as political or military resources.	AI's ability to adapt to individuals and build rapport and trust is well-documented. Conversations can reveal rich information about users' beliefs, which can in turn enable a model to appear more agreeable to the user (MIT News Office, 2026). Sustained relationship network analysis is not currently a common use case of LLMs. But it is possible that with memory improvements AI gets better at being persistent thought partners and aid in relationship network analysis. LLMs have demonstrated the ability to persuade users of false beliefs while appearing trustworthy in lab settings (Costello et al., 2026).

Risk scenario: government AI pilot	Comments on plausibility
A government agency customizes an AI assistant using region-specific training corpora (e.g., local media, policy documents, and think-tank outputs). The system increasingly reflects and reinforces locally coherent framings of international events and strategic priorities. Users interacting with similarly localized systems in other regions may experience parallel convergence toward their own regional framings. This dynamic leads to epistemic divergence across geopolitical blocs, even in the absence of deliberate propaganda or persuasion, resulting in worsened ability to reach consensus on important international issues.	Ideological segregation effects are already observed in social media contexts (González-Bailón et al., 2023). Different LLMs do have different personas and ideological biases due to various design choices (Hu et al., 2024).

Risk scenario: consumer shopping and lifestyle	Comments on plausibility
General-purpose LLMs with basic personalization features become widely available. For example, a 'memory' feature stores information that has been inferred about the user's preferences and beliefs, based on prior conversations. Users who ask questions about political candidates receive responses that agree with the inferred beliefs stored in the LLM's memory. Over time, users within each ideological cluster encounter increasingly similar framings. The system may stabilize existing epistemic trajectories, reducing opportunities for individuals to encounter information that challenges or broadens their beliefs, as illustrated by a figure below taken from (Guingrich et al., 2026).	A similar (hypothetical) scenario is described by Guingrich et al. (2026): "A shopper consults an AI for a routine grocery choice, feeling initially more 'rational', gradually relying on AI-framed norms reflected by peers, and eventually forming habitual consumption patterns that mirror large-scale homogenization of tastes".



Risk scenario: AI as public persona	Comments on plausibility
AI is increasingly used to create public personas or entities in their own right, as visible public figures (e.g., as social media influencers, singers, politicians or policy-makers). As AI public figures become widely adopted, the world appears to be populated by a diverse marketplace of ideas or opinions. However, training data, platform incentives, and commercial pressures quietly constrain the range of available thought because of underlying similarities in the AI-generated content. For example, a range of AI 'professional influencers' disagree on individual films, but converge on a shared underlying aesthetic grammar or political thought, thereby reducing consumers' access to diverse aesthetics and political ideas. Bad actors can also exploit these models to behave in specific ways that further the bad actors' interests, such as AI personas used to conduct persuasion at scale. Alternatively, bad actors use techniques to manipulate the behaviour of trusted AI personas, such as through AI grooming or jailbreaking the AI models to produce the target narratives.	AI personas described in this scenario could be generated by humans using AI tools, or envisioned as co-creators. AI-generated public personas and influencers already exist, such as Albania's AI minister for public procurements. Tracking of their views and sources is limited (Chakrabarti and Arnold, 2025; OpenArt and Fanvue, 2026; Marchal et al., 2026).

Appendix E. AI and the Widening Education Gap

While the mechanisms of cognitive offloading affect all users, their impact is not distributed equally. We discuss here how AI can entrench and widen social inequality even as it offers some leveling benefits.

Educational settings suggest AI usage will worsen existing trends in learning degradation and widening achievement gaps—and disproportionately hurt more vulnerable groups, including younger users and those with less privileged backgrounds, many of whom are in the Global South. Over the past decade, the OECD's Survey of Adult Skills (PIAAC) has documented declining literacy, numeracy, and reasoning proficiency in adults. Some studies suggest the 'Flynn effect' where a steady rise in IQ scores over the past century has stalled and may now be reversing (Dutton et al., 2016).

Isolating AI's causal impact is difficult. There are also undeniable equalizing benefits to the use of AI in education. In far too many contexts where well-resourced mentors or academic tutors are scarce, AI can serve as personalized pedagogical assistants (Burns, 2026; Marchal et al., 2026), able to adapt in real-time to the individual user's existing understanding level, speaking to them in the language they use (e.g., verbal, visual, providing links to videos and more) and potentially addressing systemic gaps in accessibility (e.g., those with various learning and sensory disabilities). AI can role-play debating to strengthen arguments and introduce perspectives that even the schooling system may not for various reasons, enhancing critical thought. AI does not tire and is not pulled away for marking and administrative duties; its system prompt is, at the very least, primed to be generally nice, which may be in unfortunate contrast to the fact that adults in children's environments can be abusive (Klimova and Pikhart, 2025).

However, these benefits exist alongside emergent evidence of cognitive path dependency in critical developmental years. We see increased AI use amongst youth aged 13-18 (National Literacy Trust, 2025), self-reported concerns by students about declining decision-making and critical thinking abilities with AI use (Malik and Sharma, 2023), and greater AI dependence associated with lower levels of critical thinking (Tian et al., 2025) and deeper cognitive engagement (Gerlich, 2025). A younger participant's quote, "I rely so much on AI that I don't think I'd know how to solve certain problems without it", illustrates this line of concern. When combined with the risks of manipulation and increased emotional dependence on AI for validation (Section 2.2), younger users still in formal education are likely to suffer motivational atrophy and disengage from real-world problem-solving. The net effect of the potential benefits and the harms will be highly context-dependent.

Nevertheless, overall, digital inequality seems to reinforce existing social inequality (United Nations Department of Economic and Social Affairs, 2020; Lythreatis et al., 2022), and AI is unlikely to be different without large-scale intervention (Ta and West, 2023). For the "Google effect", it was found that people with a larger knowledge base are less susceptible to the consequences of Internet use than those with a smaller knowledge base (Gong and Yang, 2024). For the dangers on child development from screentime use (Muppalla et al., 2023), we see clear evidence that the exposure and thus cumulative harms are stratified by income class (Common Sense Media, 2017). Perhaps unsurprisingly, then, AI's educational harms appear to fall disproportionately on marginalized students. Marginalized students fall further behind when over-relying on AI for tasks they do not fully understand, exacerbating achievement gaps and eroding problem-solving skills (Yu et al., 2025). Those with higher educational attainment are more likely to challenge AI answers before accepting them, with one study showing a 50% difference in verification rates between those with postgraduate and secondary school qualifications, perhaps due to the difference in training to be skeptical of information presented to them (Gerlich, 2025).

Some interventions designed to address cognitive offloading could also exacerbate inequality. A UNESCO global survey found that fewer than 10% of surveyed schools had any guidance on generative AI use (UNESCO, 2023). Bans or other forced reduction of usage may risk exacerbating inequalities: preventing those without available parents or guardians for homework help, or those from different language backgrounds, from accessing help. Students from well-resourced homes may learn more skill-preserving use whereas those with less support may experience AI either as prohibited or as a tool for total cognitive outsourcing, deepening post-education skill differentials in a workplace where AI is ubiquitous. This might explain the geographic difference in the survey of Claude usage by Anthropic (Anthropic, 2025c): although wealthier countries have more access to Claude, they are also less likely than lower-income countries to use it for automation—without resources like education and time, the convenience of total cognitive offloading may come at a disproportionately high cost in the long-term (Ford, 2025).

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